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Mysore
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VTU Question Papers

CIVIL ENGINEERING

Odd Semester

2021-22

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**Vidya Vikas Institute of Engineering and Technology
Bannur road, Alanahalli, Mysore-570028**

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CBCS SCHEME

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18MAT31

Third Semester B.E. Degree Examination, Feb./Mar. 2022 Transform Calculus, Fourier Series and Numerical Techniques

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

- 1** a. Evaluate (i) $L\left\{\frac{\cos 2t - \cos 3t}{0 \leq t \leq a}\right\}$ (ii) $L(t^2 e^{-3t} \sin 2t)$ (06 Marks)
- b. If $f(t) = \begin{cases} t, & 0 \leq t \leq a \\ 2a - t, & a \leq t \leq 2a \end{cases}$, $f(t + 2a) = f(t)$ then show that $L(f(t)) = \frac{1}{s^2} \tanh\left(\frac{as}{2}\right)$ (07 Marks)
- c. Solve by using Laplace Transforms
 $\frac{d^2 y}{dt^2} + 4 \frac{dy}{dt} + 4y = e^{-t}$, $y(0) = 0$, $y'(0) = 0$ (07 Marks)

OR

- 2** a. Evaluate $L^{-1}\left\{\frac{4s+5}{(s+1)^2(s+2)}\right\}$ (06 Marks)
- b. Find $L^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\}$ by using convolution theorem. (07 Marks)
- c. Express $f(t) = \begin{cases} \sin t, & 0 \leq t < \pi \\ \sin 2t, & \pi \leq t < 2\pi \\ \sin 3t, & t \geq 2\pi \end{cases}$ in terms of unit step function and hence find its Laplace Transform. (07 Marks)

Module-2

- 3** a. Obtain fourier series for the function $f(x) = |x|$ in $(-\pi, \pi)$ (06 Marks)
- b. Expand $f(x) = \frac{(\pi - x)^2}{4}$ as a Fourier series in the interval $(0, 2\pi)$ and hence deduce that
 $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$ (07 Marks)
- c. Express y as a Fourier series upto the second harmonic given :

x:	0	60	120	180	240	300
y:	4	3	2	4	5	6

(07 Marks)

OR

- 4** a. Find the Half-Range sine series of $\pi x - x^2$ in the interval $(0, \pi)$ (06 Marks)
- b. Obtain fourier expansion of the function $f(x) = 2x - x^2$ in the interval $(0, 3)$. (07 Marks)

- c. Obtain the Fourier expansion of y upto the first harmonic given :

x	0	1	2	3	4	5
y	9	18	24	28	26	20

(07 Marks)

Module-3

- 5 a. If $f(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a \end{cases}$, find the Fourier transform of $f(x)$ and hence find the value of $\int_0^{\infty} \frac{\sin x}{x} dx$ (06 Marks)

- b. Find the infinite Fourier cosine transform of $e^{-\alpha x}$. (07 Marks)

- c. Solve using z-transform $y_{n+2} - 4y_n = 0$ given that $y_0 = 0, y_1 = 2$ (07 Marks)

OR

- 6 a. Find the fourier sine transform of $f(x) = e^{-|x|}$ and hence evaluate $\int_0^{\infty} \frac{x \sin mx}{1+x^2} dx$; $m > 0$. (06 Marks)

- b. Obtain the z-transform of $\cos n\theta$ and $\sin n\theta$. (07 Marks)

- c. Find the inverse z-transform of $\frac{4z^2 - 2z}{z^3 - 5z^2 + 8z - 4}$ (07 Marks)

Module-4

- 7 a. Solve $\frac{dy}{dx} = x^3 + y$, $y(1) = 1$ using Taylor's series method considering up to fourth degree terms and find $y(1.1)$. (06 Marks)

- b. Given $\frac{dy}{dx} = 3x + \frac{y}{2}$, $y(0) = 1$ compute $y(0.2)$ by taking $h = 0.2$ using Runge - Kutta method of fourth order. (07 Marks)

- c. If $\frac{dy}{dx} = 2e^x - y$, $y(0) = 2$, $y(0.1) = 2.010$, $y(0.2) = 2.040$ and $y(0.3) = 2.090$, find $y(0.4)$ correct to 4 decimal places using Adams-Bashforth method. (07 Marks)

OR

- 8 a. Use fourth order Runge-Kutta method, to find $y(0.8)$ with $h = 0.4$, given $\frac{dy}{dx} = \sqrt{x+y}$, $y(0.4) = 0.41$ (06 Marks)

- b. Use modified Euler's method to compute $y(20.2)$ and $y(20.4)$ given that $\frac{dy}{dx} = \log_{10} \left(\frac{x}{y} \right)$ with $y(20) = 5$ Taking $h = 0.2$. (07 Marks)

- c. Apply Milne's predictor-corrector formulae to compute $y(2.0)$ given $\frac{dy}{dx} = \frac{x+y}{2}$ with

x	0.0	0.5	1.0	1.5
y	2.000	2.6360	3.5950	4.9680

(07 Marks)

Module-5

- 9 a. Using Runge-Kutta method, solve $\frac{d^2y}{dx^2} = x \left(\frac{dy}{dx} \right)^2 - y^2$, for $x = 0.2$, correct to four decimal places, using initial conditions $y(0) = 1, y'(0) = 0$ (07 Marks)
- b. Derive Euler's equation in the standard form viz, $\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$ (07 Marks)
- c. Find the extremal of the functional $\int_{x_1}^{x_2} (y^2 + y'^2 + 2ye^x) dx$ (06 Marks)

OR

- 10 a. Given the differential equation $2 \frac{d^2y}{dx^2} = 4x + \frac{dy}{dx}$ and the following table of initial values:

x	1	1.1	1.2	1.3
y	2	2.2156	2.4649	2.7514
y'	2	2.3178	2.6725	2.0657

- Compute $y(1.4)$ by applying Milne's Predictor-corrector formula. (07 Marks)
- b. Prove that geodesics of a plane surface are straight lines. (07 Marks)
- c. On what curves can the functional $\int_0^1 (y'^2 + 12xy) dx$ with $y(0) = 0, y(1) = 1$ can be extremized? (06 Marks)

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18CV32

Third Semester B.E. Degree Examination, Feb./Mar. 2022

Strength of Materials

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Derive the expression for extension of rectangular tapering bar subjected to an axial load P. (07 Marks)
- b. Explain the terms :
 (i) Modulus of Elasticity (ii) Modulus of Rigidity (iii) Poisson's Ratio. (03 Marks)
- c. A round bar with stepped portion is subjected to the forces as shown in Fig.Q1(c). Determine the magnitude of force P, such that the net deformation in the bar does not exceed 1 mm. E for steel is 200 GPa, E for Aluminium is 70 GPa. Large end diameter and small end diameter of the tapering bar are 40mm and 12.5mm respectively.

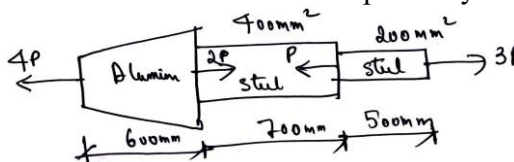


Fig.Q1(c)

(10 Marks)

OR

- 2 a. Explain the St. Venant's Principle. (04 Marks)
- b. A steel sleeve is slipped over a brass bolt and held in place by a nut. Compute the temperature rise required to stress the brass to 27.5 N/mm^2 in compression. Use the following data: $A_s = 500 \text{ mm}^2$, $A_b = 480 \text{ mm}^2$, $E_s = 19.6 \times 10^4 \text{ N/mm}^2$, $E_b = 8.82 \times 10^4 \text{ N/mm}^2$, $\alpha_s = 12 \times 10^{-6} \text{ cm/cm/}^\circ\text{C}$, $\alpha_b = 19 \times 10^{-6} \text{ cm/cm/}^\circ\text{C}$. (08 Marks)
- c. Derive the relationship between the 3 elastic constants E, G, K. (08 Marks)

Module-2

- 3 a. Show that the planes of maximum shear stresses are inclined at 45° with the principal planes. (06 Marks)
- b. The state of stress in a two-dimensionally stressed body is shown in Fig.Q3(b). Determine the principal planes, principal stresses, maximum shear stress and their planes.

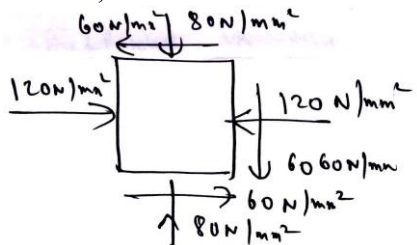


Fig.Q3(b)

(14 Marks)

OR

- 4 a. Derive Lamé's equation for radial and hoop stresses for thick cylinder subjected to internal and external fluid pressures. (08 Marks)

- b. A thick cylindrical vessel is 250 mm in internal diameter and has 50mm thick walls. It is subjected to an internal pressure of 10 MPa due to the movement of fluid. Find the maximum hoop stress developed. Also calculate the radial and hoop stress at a point 20mm from the inside surface. (08 Marks)
- c. A thin cylinder is 3m in length, 1m in diameter and has a metal thickness of 12mm in its walls. Determine the stresses (Hoop and Longitudinal) and strain along the length when subjected to an internal pressure of 1.5 MPa. Take $E = 210 \text{ GPa}$, $\mu = 0.25$. (04 Marks)

Module-3

- 5 a. For a cantilever beam subjected to a UDL of intensity $W/\text{unit length}$ throughout, plot the SFD and BMD. (06 Marks)
- b. For the beam shown in Fig.Q5(b) construct the SFD and BMD indicating salient values. Find the point of contraflexure, if any.

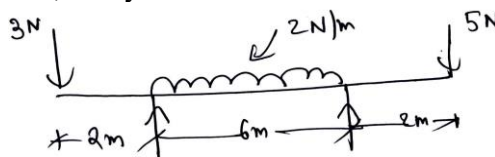


Fig.Q5(b)

(14 Marks)

OR

- 6 a. For a simply supported beam subjected to uniformly varying load of $W/\text{unit length}$ plot the SFD and BMD. (08 Marks)
- b. For the beam shown in Fig.Q6(b), find the load 'P' to have equal reactions at A and C. Draw the Bending Moment and Shear Force diagram indicating values at significant points. Locate the point of contraflexure. (12 Marks)

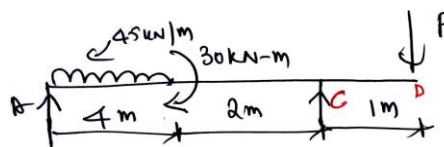


Fig.Q6(b)

Module-4

- 7 a. Define Section Modulus and Moment of Resistance. (04 Marks)
- b. Derive the relationship between Bending Stresses and Radius of curvature
ie., $\frac{\sigma}{y} = \frac{E}{R}$ (06 Marks)
- c. An unsymmetric I-section is subjected to a bending moment of 20 kN-m. The top flange being in compression. Draw the bending stress variation diagram across the section marking salient points and compute the total moment resisted by the top flange. Refer Fig.Q7(c).

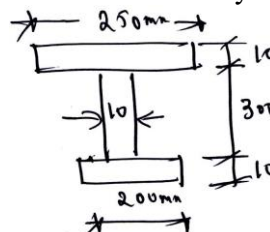


Fig.Q7(c)

(10 Marks)

OR

- 8 a. Compare the weight of solid shaft to that of the hollow shaft of the same material, having the same length to transmit power at a given speed. Take inside diameter of hollow shaft as 0.5 times the outer diameter. (10 Marks)
- c. Determine the diameter of the solid shaft which will transmit 440 kW at 280 rpm, if the maximum torsional shear stress is to be limited to 40 N/mm^2 . Assume $G = 84 \text{ kN/mm}^2$ and length of shaft is 1m with angle of twist of 1 degree. (10 Marks)

Module-5

- 9 a. For a simply supported beam subjected to an UDL of 'W' N/m determine the magnitude of maximum deflection using Double Integration method. (10 Marks)
- b. An overhanging beam ABC is loaded as shown in Fig.Q9(b). Determine the slope and deflection at its free end C. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 5 \times 10^8 \text{ mm}^4$.

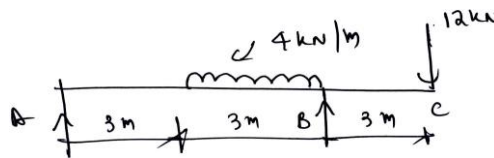


Fig.Q9(b)

(10 Marks)

OR

- 10 a. Derive the Euler's equation for buckling load on a column with one end fixed and other end hinged. (10 Marks)
- b. A hollow cast iron column whose outside diameter is 200mm has a thickness of 20mm. It is 4.5m long and is fixed at both ends. Calculate the safe load by Rankine's formula using a FOS of 4. Calculate ratio of Euler's and Rankine's critical loads for cast iron, take $\alpha = \frac{1}{1600}$, $\sigma_c = 550 \text{ N/mm}^2$, $E = 8 \times 10^4 \text{ N/mm}^2$. (10 Marks)

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18CV33

Third Semester B.E. Degree Examination, Feb./Mar. 2022 Fluids Mechanics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define fluid. Distinguish between solids, liquids and gases. (06 Marks)
- b. Define capillarity. Obtain an expression for capillary rise or drop for a given liquid of specific weight ' γ_L ' and 'd' diameter of glass tube, angle of contact is ' θ ', ' σ ' represents the surface tension force

$$h = \frac{4\sigma \cos\theta}{\gamma_L \cdot d} \quad (06 \text{ Marks})$$

- c. A 90 N rectangular solid block slides down a 30° inclined plane. The plane is lubricated by a 3 mm thick film of oil of relative density 0.90 and viscosity 0.8 Ps-sec. If the contact area is 0.3 m^2 , estimate the terminal velocity of block. (08 Marks)

OR

- 2 a. The pressure 3m below the free surface of a liquid is 13.72 kN/m^2 . Determine its specific weight and relative density. (06 Marks)
- b. Explain gauge, absolute and vacuum pressure. How do you determine the absolute pressure from the gauge pressure? (06 Marks)
- c. Find out the differential reading 'h' of an inverted U-tube manometer containing oil of specific gravity 0.7. The manometric liquid when connected across pipes 'A' and 'B' is shown in Fig.Q2(c). Convey liquids of specific gravity 1.2 and 1.0 (water) which are immiscible with manometric liquid. Pipes 'A' and 'B' are located at the same level and assume the pressures at 'A' and 'B' are equal.

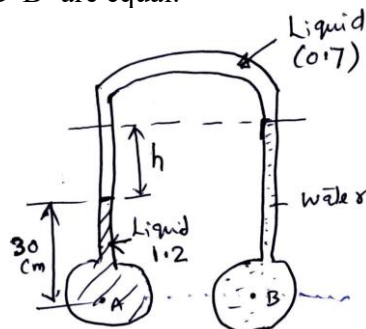


Fig.Q2(c)

(08 Marks)

Module-2

- 3 a. Define the following and mention their SI units: (i) Total Pressure (ii) Centre of pressure (iii) Total acceleration (06 Marks)
- b. An isosceles triangular plate of base 4m and altitude 4m is immersed vertically in fluid with a specific gravity 0.8. The base of the triangle is touching the top of the surface fluid horizontally and rest of its portion is within the fluid. Determine the total pressure and centre of pressure of the plate from the top liquid level. (08 Marks)
- c. Write short notes on: (i) Lagrangian method (ii) Eulerian method (iii) Flow net (06 Marks)

OR

- 4 a. A cylindrical gate is 3m long and has water on its both sides as shown in Fig.Q4(a). Determine the magnitude and direction of the resultant hydrostatic force exerted on the gate.

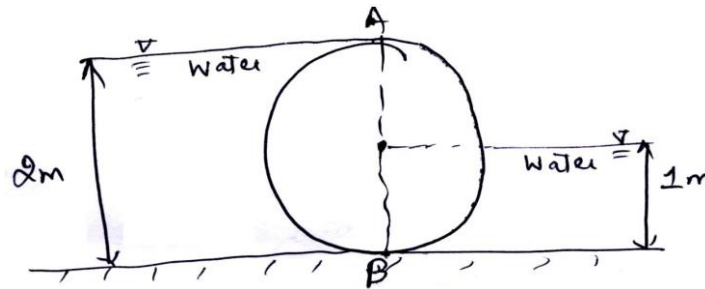


Fig.Q4(a)

(08 Marks)

- b. Distinguish between:
- Steady flow and unsteady flow
 - Rotational flow and irrotational flow
- (04 Marks)
- c. If $\phi = 3xy$, find x and y components of velocity at (1, 3) and (3, 3). Determine the discharge passing between streamlines passing through these points. (08 Marks)

Module-3

- 5 a. State and prove Euler's equation of motion and derive Bernoulli's energy equation from it. Mention the assumptions. (10 Marks)
- b. For the horizontal venturimeter of 150 mm $\times$ 75 mm, determine the reading of the mercury manometer, if the pipe carries 40 LPS of water. Given $C_d = 0.97$. Sp.Gr. mercury = 13.6. (10 Marks)

OR

- 6 a. 300 LPS of water is flowing in a pipe having diameter of 300 mm. If the pipe is bent by 120° , find the magnitude and direction of the resultant force on the bend. The pressure of the water flowing is 400 kN/m^2 . Take specific weight of water = 9.81 kN/m^3 . (08 Marks)
- b. With a neat sketch, describe the construction and working of a pitot tube. (06 Marks)
- c. An orifice meter consists of 100 mm diameter in a 300 mm diameter pipe having a coefficient of discharge of 0.65. The pipe delivers oil of relative density 0.9. The pressure difference on the two sides of the orifice plate measured by mercury differential manometer is 70 cm. Find the rate of discharge in litres per second. (06 Marks)

Module-4

- 7 a. Distinguish between the following:
- Notch and weir
 - Mouthpiece and orifice
 - Broad crested weir and sharp crested weir
 - Triangular notch and Cipolletti notch
- (08 Marks)
- b. Explain the classification of Notches and Weirs. (06 Marks)
- c. A discharge of $0.08 \text{ m}^3/\text{sec}$ was measured over a 60° angled notch. While measuring the head over notch an error of 2 mm was made. Determine the percentage error in discharge if the coefficient of discharge for the notch is 0.6. (06 Marks)

OR

- 8 a. Oil flows through a 25 mm diameter orifice under a head of 5.5 m at a rate of 3 LPS. The jet strikes a wall 1.5 m away and 120 mm vertically below the centerline of the contracted jet. Calculate the coefficients of velocity, contraction and discharge. (06 Marks)

- b. A trapezoidal notch has a base width of 0.75 m and a side slope of 1 horizontal to 2 vertical. Calculate the discharge over the notch for a head of 0.50 m by assuming $C_d = 0.63$. (06 Marks)
- c. Write a short note on ventilation of Weirs mentioning its type and effect on discharge measurement. (08 Marks)

Module-5

- 9 a. Derive an expression for the loss of head due to sudden expansion in the pipe. (08 Marks)
- b. Explain Hardy Cross method for pipe network analysis. (06 Marks)
- c. A cast iron pipe ($E = 1.0 \times 10^{11}$ Pa) is a 0.9 m in diameter and carries water ($K = 2.0 \times 10^9$ Pa) at a velocity of 2.6 m/s. A valve in this pipe is instantaneously closed bringing the flow to a sudden stop at the valve end. Estimate the water hammer head produced due to this action. The pipe thickness is 1.25 cm and the pipe can be treated as elastic. (06 Marks)

OR

- 10 a. A 6 cm diameter pipe has a discharge of water of 450 Litres/minute. At a section the pipe has a sudden expansion to a size of 9 cm diameter. If the pressure just upstream of the expansion is 20 kN/m^2 , calculate the pressure just after the expansion. Assume the pipe to be horizontal. Given $r_w = 9.81 \text{ kN/m}^3$. (06 Marks)
- b. Explain the following terms with a neat sketch:
- | | |
|-----------------------------|------------------------------|
| (i) Pipes in series | (ii) Total energy line |
| (iii) Water Hammer in pipes | (iv) Hydraulic gradient line |
- (08 Marks)
- c. The velocity of water in a 60 cm diameter and 1.5 cm thick cast iron pipe ($E = 1.04 \times 10^{11}$ Pa) is changed from 3 m/sec to zero in 0.8 sec by closure of a valve. What will be the corresponding pressure rise if given bulk modulus of elasticity of water is $2.11 \times 10^9 \text{ N/m}^2$. (06 Marks)

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18CV34

Third Semester B.E. Degree Examination, Feb./Mar. 2022 Building Materials and Construction

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What do you mean by Dressing of Stones? Explain types of dressing of stones. (08 Marks)
- b. Discuss the characteristics of good building stone. (08 Marks)
- c. Mention the qualities of good bricks. (04 Marks)

OR

- 2 a. Explain with neat sketch, the construction and working of a clamp, for burning of bricks. (08 Marks)
- b. Briefly discuss the advantages of cement concrete blocks. (06 Marks)
- c. Discuss the importance of size, shape and texture of coarse aggregates. (06 Marks)

Module-2

- 3 a. Define Foundation. Discuss functions and causes for the failure of foundation. (08 Marks)
- b. Explain different methods involved in site exploration for foundation. (06 Marks)
- c. With neat sketches discuss the features of Flemish bond and English bond of one and half thick brick wall. (06 Marks)

OR

- 4 a. Explain types of joints in stone masonry with neat sketches. (08 Marks)
- b. State the comparative merits and demerits of stone and brick masonry. (06 Marks)
- c. Describe method of construction and advantages of cavity and partition walls. (06 Marks)

Module-3

- 5 a. With neat sketches, explain various components of segmental arch. (06 Marks)
- b. Distinguish clearly between Lintel and an Arch. How does flat stone arch differs from a stone lintel? (06 Marks)
- c. Explain briefly the definition and functions of the Chejja, Balcony and Canopy. (08 Marks)

OR

- 6 a. Mention requirement of good floor, describe the method of constructing cement concrete flooring. (08 Marks)
- b. With the help of neat sketch, explain components of Queen Post Truss. (06 Marks)
- c. What are the factors to be considered while selecting roof covering? (06 Marks)

Module-4

- 7 a. Explain the following doors with neat sketches:
 - (i) Fully paneled door
 - (ii) Revolving door(06 Marks)
- b. Discuss the following windows with neat sketches:
 - (i) Bay window
 - (ii) Corner window
 - (iii) Louvered window
 - (iv) Pivoted window(08 Marks)
- c. Describe briefly classification of stairs. (06 Marks)

OR

- 8** a. Plan a dog-legged stair for a building in which the vertical distance between the floor is 3.6 meter. The stair wall measure 2.5×5.0 meter. (08 Marks)
b. What are the types of Scaffolding commonly used? (08 Marks)
c. Write explanatory note on shoring and under pinning form work. (04 Marks)

Module-5

- 9** a. Write the objectives of plastering and requirement of good plaster. (08 Marks)
b. Discuss the defects in plastering. (07 Marks)
c. Explain the method of applying stucco plastering. (05 Marks)

OR

- 10** a. Discus the causes, effects and methods of controlling dampness in building. (08 Marks)
b. Mention the objectives of painting and point out the characteristics of an ideal paint. (06 Marks)
c. Describe the procedure of painting on wood surface and inner walls of residential building. (06 Marks)

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18CV35

Third Semester B.E. Degree Examination, Feb./Mar.2022

Basic Surveying

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the classification of survey. (04 Marks)
- b. Explain the basic principles of surveying with sketches. (06 Marks)
- c. What is conventional symbols? With neat sketches show any six conventional symbols used in surveying. (10 Marks)

OR

- 2 a. What is ranging? Explain indirect ranging with neat sketch. (06 Marks)
- b. With neat sketches, explain obstacles in chaining. (08 Marks)
- c. A steel tape 20 m long standardized at 55°F with a pull of 100 N was used for measuring base line. Find the correction per tape length, if the temperature at the time of measurement was 80°F and pull exerted was 160 N. Weight of 1 cubic cm of steel is 0.0786 N. Weight of tape is 8 N and $E = 2.1 \times 10^5$ N/mm². Co-efficient of expansion of tape per 1° F = 6.2×10^{-6} . (06 Marks)

Module-2

- 3 a. Define the terms : (i) True bearing (ii) Magnetic bearing (iii) Magnetic declination and (iv) Angle of dip. (06 Marks)
- b. Differentiate between prismatic and surveyor compass (any 4). (06 Marks)
- c. The following bearings were observed in running a closed traverse:

Line	F.B.	B.B.
AB	75°5'	254°20'
BC	115°20'	296°35'
CD	165°35'	345°35'
DE	224°50'	44°5'
EA	304°50'	125°5'

At what station do you suspect the local attraction? Determine the correct magnetic bearings. If declination was 5°10' E. What are the true bearings? (08 Marks)

OR

- 4 a. Explain the Bowditch's and Transit method of adjusting closed traverse. (06 Marks)
- b. Explain closed and open traverse with neat sketch. (06 Marks)
- c. The following records were obtained in a compose traverse. Compute the length and bearing of DA:

Line	Length (m)	Bearing
AB	75.5	30°24'
BC	180.5	110°36'
CD	60.25	210°30'
DA	?	?

(08 Marks)

Module-3

- 5 a. Define the following terms:
 - (i) Back sight (ii) Fore sight (iii) Benchmark (iv) Reduced level. (06 Marks)
- b. What are the methods of leveling? Explain briefly. (06 Marks)

- c. The following consecutive readings were taken with a level and 5 m leveling staff on continuously sloping ground at a common interval of 20 m. 0.385, 1.030, 1.925, 2.825, 3.730, 4.685, 0.625, 2.005, 3.110 and 4.485 m. The reduced level of first point was 208.125 m. Rule out a page of level field book and enter the above readings. Calculate the reduced levels of points by rise and fall method and apply check. Calculate also the gradient of line joining the first and last point. (08 Marks)

OR

- 6 a. Illustrate with neat sketches:
- (i) Profile leveling
 - (ii) Differential leveling
 - (iii) Fly leveling and
 - (iv) Reciprocal leveling
- b. Enumerate the errors in leveling. (04 Marks)
- c. The following notes refer to reciprocal levels taken with one level:

Inst. at	P	Q	Remarks
P	1.824	2.748	Distance between P and Q = 1010 m
Q	0.928	1.606	RL of P = 126.386

Find :

- (i) True RL of Q
- (ii) The combined correction for curvature and refraction and
- (iii) The angular error in the collimation adjustment of the instrument. (08 Marks)

Module-4

- 7 a. Explain with neat sketch, the procedure for,
- (i) Radiation method
 - (ii) Intersection method and
 - (iii) Traversing method in plane table surveying
- b. State and explain solution to two-point problem. (08 Marks)
- c. Define Re-section. (03 Marks)

OR

- 8 a. State and explain solution to three point problem. (10 Marks)
- b. List the advantages, disadvantages and errors in plane table surveying. (10 Marks)

Module-5

- 9 a. Discuss the methods for determining areas and volume. (08 Marks)
- b. Define a contour. List the uses of contour maps. (04 Marks)
- c. The following perpendicular offsets were taken from a chain line to a hedge:

Chainage (m)	0	15	30	45	60	70	80	100	120	140
Offsets (m)	7.6	8.5	10.7	12.8	10.6	9.5	8.3	7.9	6.4	4.4

Calculate the area between survey line, the hedge and end offsets by,

- (i) Trapezoidal rule.
- (ii) Simpson's rule. (08 Marks)

OR

- 10 a. What are the characteristics of contour? Explain with sketches. (07 Marks)
- b. List the methods of contouring. Explain briefly. (05 Marks)
- c. A railway embankment 400 m long is 12 m wide at the formation level and has side slope of 2 to 1. The ground levels at every 100 m along the centre line are as under.

Distance (m)	0	100	200	300	400
R.L (m)	204.8	206.2	207.5	207.2	208.3

The formation level at zero chainage is 207.00 and the embankment has a rising gradient of 1 in 100. The ground is level across the centre line. Calculate the volume of earth work by,

- (i) Trapezoidal formula.
- (ii) Prismoidal formula. (08 Marks)

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Third Semester B.E. Degree Examination, Feb./Mar. 2022 Engineering Geology

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Using diagrams and explanation, describe the Internal Structure and Composition of Earth. (10 Marks)
 b. Write the Physical properties, Chemical composition and uses of Feldspar group of Minerals. (06 Marks)
 c. Differentiate between Rock Forming Mineral and Ore Mineral. (04 Marks)

OR

- 2 a. Discuss the relevance of Geology in Civil Engineering Profession. (08 Marks)
 b. Using appropriate examples from the Mineral Kingdom, describe the physical properties of Minerals. (08 Marks)
 c. Differentiate between Calcite and Magnetite. (04 Marks)

Module-2

- 3 a. Mention the Mineralogical composition texture, Origin and uses of
 i) Granite ii) Sandstone iii) Marble iv) Shale. (08 Marks)
 b. Explain any three types of Drainage patterns. (06 Marks)
 c. With neat sketch, explain the Soil profile. (06 Marks)

OR

- 4 a. Give a detailed account of Structure of Sedimentary Rocks. (08 Marks)
 b. Explain the types of Metamorphism. (06 Marks)
 c. Explain the types of Physical Weathering. (06 Marks)

Module-3

- 5 a. Give a detailed classification of folds and their Engineering considerations. (08 Marks)
 b. Differentiate between Horst and Graben. (06 Marks)
 c. List the various Coastline Protection Structures. (06 Marks)

OR

- 6 a. Describe the different types of the joints and mention their Engineering considerations. (08 Marks)
 b. Give the detailed account of types of Unconformities. (06 Marks)
 c. Write short note on Rock Quality Designation. (06 Marks)

Module-4

- 7 a. Give an account of vertical distribution of groundwater. (06 Marks)
 b. Write short note on Flood Control measures. (06 Marks)
 c. Give the detailed account of types of Aquifers. (08 Marks)

OR

- 8 a. Give an account of the procedure for Seismic Refraction Survey method for Groundwater Exploration. (10 Marks)
b. Give a detailed account of methods of Artificial Groundwater recharge. (10 Marks)

Module-5

- 9 a. Write the causes and effects of Earthquakes. (08 Marks)
b. Write the short note on Seismic Zones of India. (06 Marks)
c. Write a Short note on Tsunami. (06 Marks)

OR

- 10 a. Explain the application and limitations of Remote Sensing Techniques. (08 Marks)
b. Explain the components of GIS. (06 Marks)
c. Enumerate the applications of Global Positioning System. (06 Marks)

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Third Semester B.E Degree Examination, Feb./Mar. 2022

(CIVIL ENGINEERING)

COMPUTER AIDED BUILDING PLANNING AND DRAWING

Time: 3 Hours

Max. Marks: 100

NOTE:

1. Answer any *TWO* full questions from **PART A** and any *ONE* full question from **PART B**.
2. Assume any missing data suitably.

PART A

Q1	A one way slab for a hall of internal dimension 7.0×11.77m has the following details. Thickness of slab = 150mm Wall Thickness = 230mm Main steel along short span = 10 mm # @ 100mm c/c Distribution steel = 8mm # @ 150mm c/c Draw to suitable scale the following a. Plan showing the reinforcement details. b. Cross section of slab at mid span along short span. c. Cross section of slab at mid span along long span. (25 Marks)
Q2	The scale the elevation and cross section of English bond and Flemish bond with all the details for 12 courses. (25 Marks)
Q3	Draw the plan and Section elevation for a septic tank for the following details. Depth of tank= 1.75m, Length of PCC bed=4.7m, width of PCC bed 1.9m, Thickness of PCC bed 0.15m. Width of tank wall in brick work above PCC bed=0.4m for a height of 0.4m. Width of tank wall in brick work=0.3m for a height of 0.5m. Width of tank wall in brick work=0.2m for a height of 0.7m. The Tank consists of a RCC pre cast slab of thickness 7.5cm. Also show the provision for inlet and outlet pipes (25 Marks)
Q4	Draw to scale the plan and sectional elevation of both the flight of a open navel stair with rectangular well for an office building with the following data. Inside dimension of staircase = 6× 4.5m, Height between the floors = 3.75m, Thickness of the floor slab and the landing slab =150mm, Width of stair =1.5m (25 Marks)

PART B

Q5	The line diagram of a residential building is given in Fig. Q5. Draw to scale the following: a) Plan at sill level. b) Front elevation. c) Section along XX. d) Schedule of openings. (50 Marks)
Q6	The line diagram of Residential building is given in Fig. Q6. Draw to scale the following: a) Plan at sill level. b) Front elevation. c) Section along XX. d) Schedule of openings. (50 Marks)

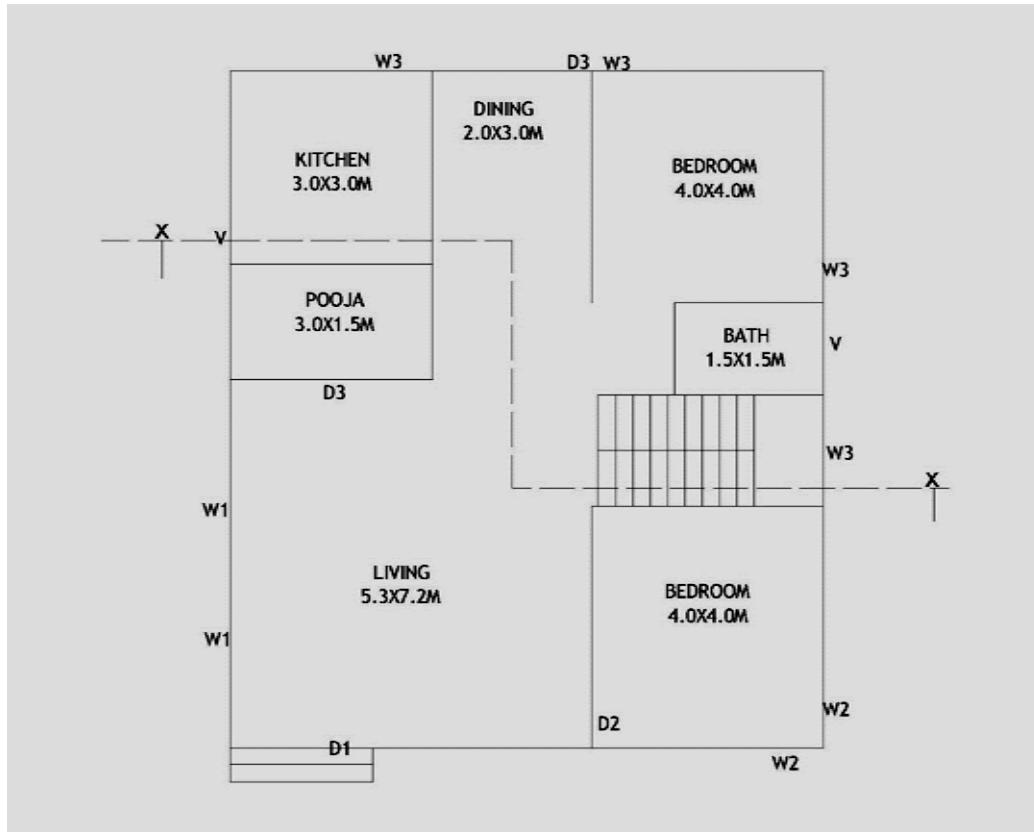


Fig. Q5

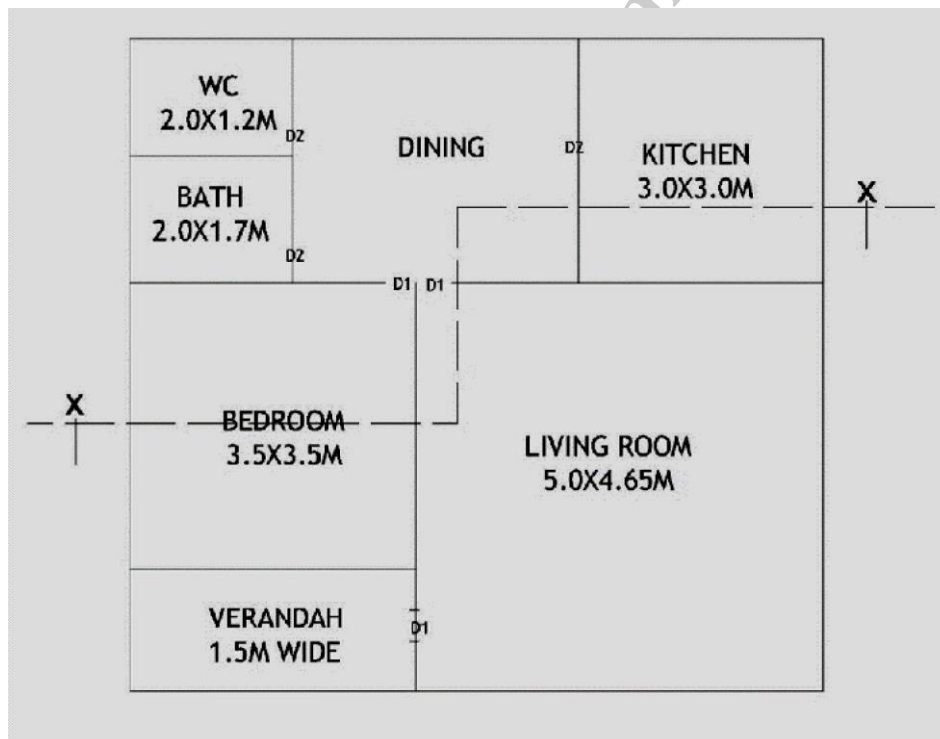


Fig. Q6

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Question Paper Version : A

Third/Fourth Semester B.E. Degree Examination, Feb./Mar. 2022
Constitution of India, Professional Ethics and Cyber Law
(COMMON TO ALL BRANCHES)

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the hundred questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. The Constitution of India was enacted by a Constituent Assembly set up
 - a) Under the Cabinet Mission Plan 1946
 - b) Under the Indian Independence Act 1947
 - c) Under a resolution of the Provisional Government.
 - d) By the Indian National Congress.
 2. The Members of the Constituent Assembly are
 - a) Directly elected by the people
 - b) Nominated by various Political Parties
 - c) Nominated by rulers of the Indian States
 - d) Elected by the Provincial Assemblies
 3. The federal feature of the Indian Constitution provides for
 - a) Distribution of Legislative powers between the Union Government and the State Government.
 - b) Division of powers between the Executive and Judiciary.
 - c) Distribution of powers between the P.M and Cabinet.
 - d) None of these
 4. The Governor of State is
 - a) Directly elected by the people
 - b) Elected by the State Legislature
 - c) Appointed by the President
 - d) Nominated by the Parliament.
 5. The source of Authority of the Indian Constitution is
 - a) The Government of India
 - b) The People of India
 - c) The President
 - d) The Parliament
 6. The Preamble was amended by
 - a) 24th Amendment
 - b) 42nd Amendment
 - c) 39th Amendment
 - d) None of the above

7. Fraternity means
 - a) Spirit of brotherhood
 - b) Fatherly treatment
 - c) Unity and integrity
 - d) Elimination of Economic Justice
8. In the final form of the Constitution adopted by the Constituent Assembly, how many Articles and Schedules were there?
 - a) 397 Articles and 7 Schedules
 - b) 395 Articles and 4 Schedules
 - c) 400 Articles and 10 Schedules
 - d) 395 Articles and 8 Schedules
9. The Preamble of the Indian Constitution does not contain concept of
 - a) Democratic
 - b) Adult Franchise
 - c) Sovereignty
 - d) Fraternity
10. The strength of the Constituent Assembly, after the withdrawal of the Muslim League, was reduced to
 - a) 299 members
 - b) 329 member
 - c) 331 members
 - d) 359 members
11. Who of the following acted as the Constitutional Advisor of the Constituent Assembly
 - a) Dr. B.R. Ambedkar
 - b) Dr. Babu Rajendra Prasad
 - c) B.N. Rao
 - d) Dr. Sachidanand Sinha
12. Which one of the following provisions of the Constitution came into force soon after its adoption on 26th November 1949?
 - a) Provision relating to Citizenship
 - b) Elections
 - c) Provisional Parliament
 - d) All the above
13. The three types of Justice referred in our Preamble are
 - a) Social, Economic and Social
 - b) Economic, International and Political
 - c) Economic, Religious and Social
 - d) Religious, Social and Political
14. What was the exact Constitutional status of the Indian Republic on January 26, 1950, when the Constitution was inaugurated?
 - a) A Democratic Republic
 - b) Sovereign Democratic Republic
 - c) A Sovereign Secular Democratic Republic
 - d) A Sovereign Socialist Secular Democratic Republic.
15. Right to against Exploitation seeks to protect the weaker sections of Society by
 - a) Giving equal pay for equal work for men and women.
 - b) Prohibiting human trafficking and beggar
 - c) Providing compulsory education for children below the age of 14 years
 - d) Forcing a person to work against his will without payment
16. Which one of the following Directive principles can be described as Gandhian in nature?
 - a) Providing equal pay for equal work for both Men and Women
 - b) Workers participation in Management
 - c) Organization of Village Panchayats as units of self Government
 - d) Separation of Judiciary from the Executive.
17. Who has been vested with the power to decide whether the restrictions imposed on the Fundamental Rights of Indian Citizen are reasonable or not
 - a) The Parliament
 - b) The President
 - c) The Courts
 - d) None of the above

18. Which one of the following Rights conferred by the Constitution is also available to Noncitizens,
- Freedom of speech assembly and association
 - Freedom to move, reside and settle in any part of the territory of India
 - Freedom to acquire property or to carry on any occupation, trade or business
 - Right to Constitutional remedies.
19. Which one of the following has been wrongly listed as a special feature of Fundamental Rights in India
- Fundamental Rights are more sacrosanct than rights granted by ordinary laws
 - Fundamental Rights are subject to reasonable restrictions
 - Fundamental Rights are Justifiable and can be enforced through the Supreme Court
 - None of these.
20. The main objective of the Cultural and Educational Rights granted to the Citizen is
- To preserve the rich culture heritage of India.
 - To evolve a single integrated India culture.
 - To help the minorities to conserve their culture.
 - All the above.
21. The Fundamental Rights of a Citizen can be suspended
- By the Parliament through a Law enacted by two – third majority
 - By the President during a National emergency
 - By the Supreme Court
 - None of these
22. Which authority can a Citizen approach for securing right of Personal freedom
- The Parliament
 - The President
 - Supreme Court alone
 - Both Supreme Court and High Court
23. The main objective of the Fundamental Rights is to
- Ensure Independence of Judiciary
 - Promote a Socialist Pattern of Society
 - Ensure Individual liberty
 - Ensure all the above
24. Under which section of IT Act, stealing any digital asset or information is written a cyber crime
- 65
 - 65 - D
 - 67
 - 70
25. Fundamental duties of the Indian Citizen, were
- Enshrined in the original Constitution
 - Added to the Constitution by the 42nd Amendment.
 - Added to the Constitution by the 44th Amendment.
 - Added to the Constitution in the wake the Supreme Court Judgment Keshavananda Bharati case with consent of all the Political parties.
26. Which one of the following Fundamental Right has been subject of maximum litigation since the inauguration of the Constitution?
- Right to Freedom of speech
 - Right to Constitutional remedies
 - Right to property
 - Right against exploitation
27. The Fundamental Rights of Citizens were
- Incorporated in the original Constitution
 - Outlined in an Act of Parliament in 1952
 - Incorporated by the 42nd Amendment
 - Incorporated by the 44th Amendment

28. The Fundamental Rights of Indian Citizen have been criticized on the ground that
a) They are hemmed in by too many restrictions.
b) They are couched in language beyond the comprehension of ordinary citizen
c) They are absolute
d) Both (a) and (b).
29. Respite means
a) Death due to drowning
b) Awarding lesser punishment
c) Death due to strangulation
d) Painless death
30. The Governor recommends the imposition of Presidents rule in the State
a) On the recommendation of the State Legislature
b) On the recommendation of the C.M.
c) On the recommendation of Council of Minister
d) If he is satisfied that the State Government cannot be carried on his accordance with the provision of the Constitution.
31. The Chief Election Commissioner can be removed from his office before the expiry of term by the
a) Chief Justice of India
b) Prime Minister on the recommendation of Cabinet
c) President on the recommendation of Parliament after the impeachment
d) President on the advice of Chief Justice of India.
32. The quorum of minimum number of members required to hold the meetings of either Houses of Parliament is
a) One - tenth
b) One - fifth
c) One - third
d) 72 hours
33. The advice of the Supreme Court is
a) Binding of the President
b) Binding on the President if it is tendered unanimously
c) Not binding on the President
d) Binding in certain cases and not binding in other cases
34. The Governor reserves the Right to issue ordinances
a) When the State Legislature is not in session and he feels that there is an immediate need of action
b) Whenever the State is under President's Rule
c) Whenever he likes
d) None of these
35. The Writ of Certiorari is issued by a Superior Court
a) To an Inferior Court to stop further proceedings in a particular case
b) To an Inferior Court to transfer the record of proceedings in a case for its review
c) To an Office to show his right to hold a particular Office
d) To a Public authority to produce a person detained by if before the Court within 24 hours.
36. Which one of the following was wrongly listed as a duty of Indian Citizens
a) To uphold and protect the Sovereign unity and Integrity of the Country
b) To promote harmony and the spirit of common brotherhood among the people of India
c) To protect and pressure the Natural Environment
d) To practice Family planning and control population.

37. The Directive Principles Aim at
a) Ensuring Individual liberty
b) Ensuring strengthening of the Country's Independence
c) Providing a social and economic base for a genuine Democracy in the Country.
d) Achieving all the above objectives.
38. The Directive Principles are the
a) Positive instructions to the Government to work for the attainment of the set objectives
b) Negative injunctions to the Government to refrain from encroaching on the freedom of the people.
c) Directive to the State to enhance the International prestige of the Country
d) Directives to the Government to pursue a policy of non alignment.
39. Which one of the following has been wrongly listed as Directive Principle based on 'Liberal Principles'?
- a) Separation of Judiciary and Executive
b) Provision of a Uniform Civil code for the Country
c) Protection of monuments and places of artistic or Historical importance
d) None of the above has been wrongly listed.
40. The Constitution has vested the Executive power of the Union Government in
a) The President of India
b) The Prime Minister
c) The Council of Minister
d) All the above
41. Which one of the following does not take part in the Election of the President?
- a) Elected members of Lok - Sabha
b) Elected members of Rajya – Sabha
c) Members of the Legislative Council
d) None of these
42. The President can be removed by impeachment procedure on the ground of violating the Constitution by
a) The Supreme Court
b) The Lok – Sabha only
c) Both Houses of Parliament
d) The High Court
43. The Vice – President of India is Elected by the
a) People
b) Members of State – Legislative Assembly
c) Members of the Rajya – Sabha
d) Members of both the Houses of Parliament at Joint sitting.
44. Who discharges the duties of the President in the event of President and Vice – President being not available?
- a) The Prime Minister
b) The Chief Justice of India
c) The Speaker of Lok - Sabha
d) The Attorney General of India
45. Which one of the following can the President of India declare?
- a) Emergency due to threat of War, external aggression or armed rebellion
b) Emergency due to breakdown of constitutional machinery in the State
c) Financial emergency on account of threat to the financial credit of India
d) All the above.
46. Which budget is placed first in the Parliament House
a) Railway
b) General budget
c) Financial
d) Vote of credit

47. The President can make Laws through ordinances
- During the recess of the Parliament
 - On certain subjects even when Parliament is in session
 - Only on subjects contained in the concurrent list
 - Under no circumstances.
48. The President can grant pardon in
- All cases of punishment by Court martial
 - All offences against laws in the Union and Concurrent list
 - All cases involving death sentence
 - All the above cases
49. If State fails to comply with the directives of the Central Government, the President can
- Declare break – down of Constitutional machinery in the State and assume responsibility for its governance
 - Send reserve police force to secure compliance with directions
 - Dissolve the State legislature and order fresh elections
 - Can do either (a) or (b)
50. Which one of the following has been wrongly listed as Judicial power of the President of India?
- He appoints the Chief Justice and other Judges of the Supreme Court
 - He can grant pardon, reprieve and respite to a person awarded punishment
 - He can consult the Supreme Court on any question of law or fact.
 - He can remove the Judges of Supreme – Court on ground of misconduct.
51. Impeachment proceedings can be initiated against the President in either House of Parliament only if a resolution signed by members of the house is moved.
- 10 percent of total
 - 25 percent of total
 - 20 percent of total
 - 15 percent of total
52. Which one of the following functions of Prime – Minister has been wrongly listed?
- He presides over the meeting of the Cabinet
 - He prepares the agenda for the meeting of the Cabinet.
 - He coordinates the working of various department
 - He chairs the meeting of the various standing and ad-hoc committees of Parliament.
53. A motion of no – confidence against the Council Ministers can be moved in the Lok – Sabha , if it is supported by atleast
- 50 members
 - 55 members
 - 100 members.
 - One – third of the total members of Lok - Sabha
54. The President can call a Joint Session of the two Houses of Parliament
- If a bill passed by one house is rejected by the other
 - If the amendment proposed to the bill by one house is not acceptable to the other house.
 - If the house does not take any action for six months on a bill remitted by the other house.
 - Under all the above conditions.

55. The Members of the Rajya – Sabha except the nominated ones are
a) Directly elected by the people b) Elected by local Self – Governing bodies
c) Elected by the Legislative Assemblies of the States.
d) Elected partially by Legislative Assemblies and partially by the Local Self Governing bodies.
56. The President who is the head of the State under the Parliamentary system prevailing in India.
a) Enjoys absolute powers b) Enjoys limited but real powers
c) Enjoys only nominal powers d) Enjoys no powers
57. The Vice – President is the Ex – Office Chairman of
a) The Rajya Sabha b) The National Development Council
c) The Planning Commission d) None of the above
58. The Speaker of the Lok – Sabha is
a) Appointed by the President.
b) Appointed by the President on the recommendations of the P.M.
c) Elected by the members of the two houses at a joint sitting.
d) Elected by the members of the Lok – Sabha.
59. The Rajya – Sabha is a permanent House but
a) One – third of its members retire every two years
b) One – half of its members retire every three years
c) One – fifth of its members retire every year
d) One – half of its member retire every two years.
60. The power to control the Expenditure of the Government of India rests exclusively with
a) The Parliament b) The President
c) The Comptroller and Auditor General d) The Union Finance Minister
61. Who decides disputes regarding disqualification of members of Parliament?
a) The President b) The Concerned house
c) The Election Commission.
d) The President in consultation with the Election Commission.
62. Who presides over the Lok – Sabha if neither the Speaker nor the Deputy Speaker is not available?
a) A member nominated by the President.
b) A member chosen by the Council of Minister.
c) A member of the panel of Chairman announced by the Speaker.
d) The Senior most member of the Lok – Sabha.
63. Lok – Sabha is superior to the Rajya – Sabha because
a) It is directly elected b) It alone controls the Finances
c) It can oust the Council of Minister through a Vote of no – Confidence
d) of all the above reasons.
64. The Supreme Court of India was setup
a) By the Constitution b) Under the Indian Independence Act 1947
c) Through an Act of Parliament in 1950 d) Under the Government of India Act 1935

65. The Judges of the Supreme Court are
 a) Elected by the Parliament.
 b) Appointed by the President on the advice of the Parliament.
 c) Appointed by the President on the advice of the P.M.
 d) Appointed by the President on the advice of the Chief Justice of India.
66. The Judges of the Supreme Court after retirement are not permitted to carry on practice before
 a) The Supreme Court
 b) The High Courts
 c) The District and Session Courts
 d) Any of the above
67. Which of the following Jurisdiction of the Supreme Court of India has been wrongly listed
 a) Original Jurisdiction
 b) Appellate Jurisdiction
 c) Advisory Jurisdiction
 d) None of the above
68. Generally, the Governor belongs to
 a) The State where he is posted
 b) Some other State
 c) The Indian Administrative Service
 d) None of the above
69. Which of the following Legislative Powers is enjoyed by the Governor of a State?
 a) He can summon or prorogue the State Legislature
 b) He can appoint one sixth of the members of the Legislative Council.
 c) He can nominate certain member of the Anglo Indian Community to the Legislative Assembly.
 d) All of above powers.
70. Engineering Ethics is a
 a) Preventive Ethics
 b) Developing
 c) Natural Ethics
 d) Scientifically developed Ethics.
71. Cooking Means
 a) Boiling under pressure
 b) Retaining results which fit theory
 c) Making deceptive statements
 d) Misleading the Public about quality of the product.
72. Which one is not a Trade Secret?
 a) Theorem
 b) Equipment
 c) Formulae
 d) Pattern
73. The codes of Ethics can be taken as guidelines by the Engineers to
 a) Resolve the conflicts
 b) Formulate problems
 c) Overcome the work pressure
 d) Escape from the responsibility
74. A Fault tree is used to
 a) Assess the risk involved
 b) Claim compensation
 c) Take free consent
 d) Improve safety
75. Risk of harm equal to probability of producing benefit is
 a) Inevitable Risk
 b) Acceptable Risk
 c) Risk which cannot be avoided
 d) None of these

76. One of the Aims of studying Engineering Ethics is to
a) Inspire Engineers to acquire in depth knowledge in their field
b) Acquire new skills in Engineering testing.
c) Stimulate moral imaginations
d) Make Engineers self – confident in discharging their duties.
77. Which one is not an impediment to responsibility?
a) Group think
b) Microscopic vision
c) Trademark
d) Egocentric tendencies
78. Being safe or blaming others is type of attitude of responsibility of Engineers
a) Minimalist
b) Reasonable care
c) Good works
d) None of these
79. To overcome an impediment ‘Uncritical Acceptance’, what step an Engineer has to take?
a) Accept and Analyse
b) Analyse and Accept
c) Always say Yes Boss
d) None of these
80. Engineering Ethics ,
a) stimulates the moral imagination
b) Provides up – to – date knowledge in the field of Engineering.
c) Stimulates to Conduct Research
d) Stresses on Time Management.
81. Professional Ethics is
a) Set of Rules relating to personal character of Professionals
b) Traditional Rules observed since a long time.
c) Set of Rules passed by Professional bodies.
d) Set of standards adopted by Professionals.
82. Tight couple means
a) Binding two beams tightly
b) Erecting two pillars side by side
c) Process tightly coupled
d) Strong adhesive material
83. An Engineer may not be held legally liable or causing harm. When the harm is caused
a) Intentionally
b) Ignorantly
c) Negligently
d) Recklessly
84. A compound measure of the probability and magnitude of the adverse effect is known as.
a) Risk
b) Benefit
c) Compensation
d) Both (b) and (c)
85. Engineers shall issue Public statements only
a) In subjective manner
b) In objective manner
c) On their personal responsibility
d) Based on the reports sent by higher Officers.
86. Attackers commonly target _____ for fetching IP address of a target or victim user.
a) Website
b) Web pages
c) IP tracker
d) Emails
87. _____ is the first phase of Ethical hacking
a) DNS Poisoning
b) Foot printing
c) ARP – Poisoning
d) Enumeration
88. Which of the following do not comes under the intangible skills of hackers?
a) Creative thinking
b) Problem solving capability
c) Persistence
d) Smart attacking potential

89. Why programming language is important for ethical hackers and Security Professionals?
a) Only to write malware.
b) For solving problems and building tool and programs
c) To teach programming
d) To develop program to harm others.
90. Understanding of _____ is also important for gaining access to a system through networks.
a) OS b) Email servers c) Networking d) Hardware
91. For hacking a database or accessing and manipulating data which of the following language the hacker must know?
a) SQL b) HTML c) TCL d) F #
92. _____ are piece of programs or scripts that allow hackers to take control over any system.
a) Exploits b) Antivirus c) Firewall by passers d) Worms
93. The process of finding vulnerabilities and exploiting them using exploitable scripts or programs are known as
a) Infiltrating b) Exploitation c) Cracking d) Hacking
94. How many types of exploits are there based on their nature from hackings perspective?
a) 04 b) 03 c) 02 d) 05
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a) Phishing b) Injecting Trojans to a target victim
c) MITM d) Credit card details leak in deep web
100. In which year India's IT Act came into existence?
a) 2000 b) 2001 c) 2002 d) 2003

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Question Paper Version : B

Third/Fourth Semester B.E. Degree Examination, Feb./Mar. 2022
Constitution of India, Professional Ethics and Cyber Law
(COMMON TO ALL BRANCHES)

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the hundred questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
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5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Which one of the following does not take part in the Election of the President?
 a) Elected Members of Lok - Sabha b) Elected Members of Rajya – Sabha
 c) Members of the Legislative Council d) None of these
 2. The President can be removed by impeachment procedure on the ground of violating the Constitution by
 a) The Supreme Court b) The Lok – Sabha only
 c) Both Houses of Parliament d) The High Court
 3. The Vice – President of India is Elected by the
 a) People
 b) Members of State – Legislative Assembly
 c) Members of the Rajya – Sabha
 d) Members of both the Houses of Parliament at Joint sitting.
 4. Who discharges the duties of the President in the event of President and Vice – President being not available?
 a) The Prime Minister b) The Chief Justice of India
 c) The Speaker of Lok - Sabha d) The Attorney General of India
 5. Which one of the following can the President of India declare?
 a) Emergency due to threat of War, external aggression or armed rebellion
 b) Emergency due to breakdown of constitutional machinery in the State
 c) Financial emergency on account of threat to the financial credit of India
 d) All the above.
 6. Which budget is placed first in the Parliament House
 a) Railway b) General budget c) Financial d) Vote of credit

7. The President can make Laws through ordinances
 - a) During the recess of the Parliament
 - b) On certain subjects even when Parliament is in session
 - c) Only on subjects contained in the concurrent list
 - d) Under no circumstances.
8. The President can grant pardon in
 - a) All cases of punishment by Court martial
 - b) All offences against laws in the Union and Concurrent list
 - c) All cases involving death sentence
 - d) All the above cases
9. If State fails to comply with the directives of the Central Government, the President can
 - a) Declare break – down of Constitutional machinery in the State and assume responsibility for its governance
 - b) Send reserve police force to secure compliance with directions
 - c) Dissolve the State legislature and order fresh elections
 - d) Can do either (a) or (b)
10. Which one of the following has been wrongly listed as Judicial power of the President of India?
 - a) He appoints the Chief Justice and other Judges of the Supreme Court
 - b) He can grant pardon, reprieve and respite to a person awarded punishment
 - c) He can consult the Supreme Court on any question of law or fact.
 - d) He can remove the Judges of Supreme – Court on ground of misconduct.
11. Who decides disputes regarding disqualification of Members of Parliament?
 - a) The President
 - b) The Concerned house
 - c) The Election Commission.
 - d) The President in consultation with the Election Commission.
12. Who presides over the Lok – Sabha if neither the Speaker nor the Deputy Speaker is not available?
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 - a) By the Constitution
 - b) Under the Indian Independence Act 1947
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16. The Judges of the Supreme Court after retirement are not permitted to carry on practice before
 - a) The Supreme Court
 - b) The High Courts
 - c) The District and Session Courts
 - d) Any of the above
17. Which of the following Jurisdiction of the Supreme Court of India has been wrongly listed
 - a) Original Jurisdiction
 - b) Appellate Jurisdiction
 - c) Advisory Jurisdiction
 - d) None of the above
18. Generally, the Governor belongs to
 - a) The State where he is posted
 - b) Some other State
 - c) The Indian Administrative Service
 - d) None of the above
19. Which of the following Legislative Powers is enjoyed by the Governor of a State?
 - a) He can summon or prorogue the State Legislature
 - b) He can appoint one sixth of the members of the Legislative Council.
 - c) He can nominate certain member of the Anglo Indian Community to the Legislative Assembly.
 - d) All of above powers.
20. Engineering Ethics is a
 - a) Preventive Ethics
 - b) Developing
 - c) Natural Ethics
 - d) Scientifically developed Ethics.
21. Professional Ethics is
 - a) Set of Rules relating to personal character of Professionals
 - b) Traditional Rules observed since a long time.
 - c) Set of Rules passed by Professional bodies.
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27. _____ is the first phase of Ethical hacking
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 - b) Foot printing
 - c) ARP – Poisoning
 - d) Enumeration

28. Which of the following do not comes under the intangible skills of hackers?
 - a) Creative thinking
 - b) Problem solving capability
 - c) Persistence
 - d) Smart attacking potential
29. Why programming language is important for ethical hackers and Security Professionals?
 - a) Only to write malware.
 - b) For solving problems and building tool and programs
 - c) To teach programming
 - d) To develop program to harm others.
30. Understanding of _____ is also important for gaining access to a system through networks.
 - a) OS
 - b) Email servers
 - c) Networking
 - d) Hardware
31. The Constitution of India was enacted by a Constituent Assembly set up
 - a) Under the Cabinet Mission Plan 1946
 - b) Under the Indian Independence Act 1947
 - c) Under a resolution of the Provisional Government.
 - d) By the Indian National Congress.
32. The Members of the Constituent Assembly are
 - a) Directly elected by the people
 - b) Nominated by various Political Parties
 - c) Nominated by rulers of the Indian States
 - d) Elected by the Provincial Assemblies
33. The Federal feature of the Indian Constitution provides for
 - a) Distribution of Legislative powers between the Union Government and the State Government.
 - b) Division of powers between the Executive and Judiciary.
 - c) Distribution of powers between the P.M and Cabinet.
 - d) None of these
34. The Governor of State is
 - a) Directly Elected by the people
 - b) Elected by the State Legislature
 - c) Appointed by the President
 - d) Nominated by the Parliament.
35. The source of Authority of the Indian Constitution is
 - a) The Government of India
 - b) The People of India
 - c) The President
 - d) The Parliament
36. The Preamble was amended by
 - a) 24th Amendment
 - b) 42nd Amendment
 - c) 39th Amendment
 - d) None of the above
37. Fraternity means
 - a) Spirit of brotherhood
 - b) Fatherly treatment
 - c) Unity and integrity
 - d) Elimination of Economic Justice
38. In the final form of the Constitution adopted by the Constituent Assembly, how many Articles and Schedules were there?
 - a) 397 Articles and 7 Schedules
 - b) 395 Articles and 4 Schedules
 - c) 400 Articles and 10 Schedules
 - d) 395 Articles and 8 Schedules

39. The Preamble of the Indian Constitution does not contain concept of
a) Democratic b) Adult Franchise c) Sovereignty d) Fraternity
40. The strength of the Constituent Assembly, after the withdrawal of the Muslim League, was reduced to
a) 299 members b) 329 member c) 331 members d) 359 members
41. The Fundamental Rights of a Citizen can be suspended
a) By the Parliament through a Law enacted by two – third majority
b) By the President during a National emergency
c) By the Supreme Court d) None of these
42. Which authority can a Citizen approach for securing Right of Personal freedom
a) The Parliament b) The President
c) Supreme Court alone d) Both Supreme Court and High Court
43. The main objective of the Fundamental Rights is to
a) Ensure Independence of Judiciary b) Promote a Socialist Pattern of Society
c) Ensure Individual liberty d) Ensure all the above
44. Under which section of IT Act, stealing any digital asset or information is written a cyber crime
a) 65 b) 65 - D c) 67 d) 70
45. Fundamental duties of the Indian Citizen, were
a) Enshrined in the original Constitution
b) Added to the Constitution by the 42nd Amendment.
c) Added to the Constitution by the 44th Amendment.
d) Added to the Constitution in the wake the Supreme Court Judgment Keshavananda Bharati case with consent of all the Political parties.
46. Which one of the following Fundamental Right has been subject of maximum litigation since the Inauguration of the Constitution?
a) Right to Freedom of Speech b) Right to Constitutional Remedies
c) Right to Property d) Right against Exploitation
47. The Fundamental Rights of Citizens were
a) Incorporated in the original Constitution b) Outlined in an Act of Parliament in 1952
c) Incorporated by the 42nd Amendment d) Incorporated by the 44th Amendment
48. The Fundamental Rights of Indian Citizen have been criticized on the ground that
a) They are hemmed in by too many restrictions.
b) They are couched in language beyond the comprehension of ordinary citizen
c) They are absolute d) Both (a) and (b).
49. Respite means
a) Death due to drowning b) Awarding lesser punishment
c) Death due to strangulation d) Painless death
50. The Governor recommends the imposition of Presidents rule in the State
a) On the recommendation of the State Legislature
b) On the recommendation of the C.M.
c) On the recommendation of Council of Minister
d) If he is satisfied that the State Government cannot be carried on his accordance with the provision of the Constitution.

51. Who of the following acted as the Constitutional Advisor of the Constituent Assembly

 - a) Dr. B.R. Ambedkar
 - b) Dr. Babu Rajendra Prasad
 - c) B.N. Rao
 - d) Dr. Sachidanand Sinha
52. Which one of the following provisions of the Constitution came into force soon after its adoption on 26th November 1949?

 - a) Provision relating to Citizenship
 - b) Elections
 - c) Provisional Parliament
 - d) All the above
53. The three types of Justice referred in our Preamble are

 - a) Social, Economic and Social
 - b) Economic, International and Political
 - c) Economic, Religious and Social
 - d) Religious, Social and Political
54. What was the exact Constitutional status of the Indian Republic on January 26, 1950, when the Constitution was inaugurated?

 - a) A Democratic Republic
 - b) Sovereign Democratic Republic
 - c) A Sovereign Secular Democratic Republic
 - d) A Sovereign Socialist Secular Democratic Republic.
55. Right to against Exploitation seeks to protect the weaker sections of Society by

 - a) Giving equal pay for equal work for men and women.
 - b) Prohibiting human trafficking and beggar
 - c) Providing compulsory education for children below the age of 14 years
 - d) Forcing a person to work against his will without payment
56. Which one of the following Directive principles can be described as Gandhian in nature?

 - a) Providing equal pay for equal work for both Men and Women
 - b) Workers participation in Management
 - c) Organization of Village Panchayats as units of self Government
 - d) Separation of Judiciary from the Executive.
57. Who has been vested with the power to decide whether the restrictions imposed on the Fundamental Rights of Indian Citizen are reasonable or not

 - a) The Parliament
 - b) The President
 - c) The Courts
 - d) None of the above
58. Which one of the following Rights conferred by the Constitution is also available to Noncitizens,

 - a) Freedom of speech assembly and association
 - b) Freedom to move, reside and settle in any part of the territory of India
 - c) Freedom to acquire property or to carry on any occupation, trade or business
 - d) Right to Constitutional remedies.
59. Which one of the following has been wrongly listed as a special feature of Fundamental Rights in India

 - a) Fundamental Rights are more sacrosanct than rights granted by ordinary laws
 - b) Fundamental Rights are subject to reasonable restrictions
 - c) Fundamental Rights are Justifiable and can be enforced through the Supreme Court
 - d) None of these.

60. The main objective of the Cultural and Educational Rights granted to the Citizen is
a) To preserve the rich culture heritage of India.
b) To evolve a single integrated India culture.
c) To help the minorities to conserve their culture.
d) All the above.
61. For hacking a database or accessing and manipulating data which of the following language the hacker must know?
a) SQL b) HTML c) TCL d) F #
62. _____ are piece of programs or scripts that allow hackers to take control over any system.
a) Exploits b) Antivirus c) Firewall bypassers d) Worms
63. The process of finding vulnerabilities and exploiting them using exploitable scripts or programs are known as
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a) Phishing b) Injecting Trojans to a target victim
c) MITM d) Credit card details leak in deep web
70. In which year India's IT Act came into existence?
a) 2000 b) 2001 c) 2002 d) 2003
71. The Chief Election Commissioner can be removed from his office before the expiry of term by the
a) Chief Justice of India
b) Prime Minister on the recommendation of Cabinet
c) President on the recommendation of Parliament after the impeachment
d) President on the advice of Chief Justice of India.
72. The quorum of minimum number of members required to hold the meetings of either houses of Parliament is
a) One - tenth b) One - fifth c) One - third d) 72 hours

73. The advice of the Supreme Court is
a) Binding of the President
b) Binding on the President if it is tendered unanimously
c) Not binding on the President
d) Binding in certain cases and not binding in other cases
74. The Governor reserves the Right to issue ordinances
a) When the State Legislature is not in session and he feels that there is an immediate need of action
b) Whenever the State is under President's Rule
c) Whenever he likes
d) None of these
75. The Writ of Certiorari is issued by a Superior Court
a) To an Inferior Court to stop further proceedings in a particular case
b) To an Inferior Court to transfer the record of proceedings in a case for its review
c) To an Office to show his right to hold a particular Office
d) To a Public authority to produce a person detained by it before the Court within 24 hours.
76. Which one of the following was wrongly listed as a duty of Indian Citizens
a) To uphold and protect the Sovereign unity and integrity of the Country
b) To promote harmony and the spirit of common brotherhood among the people of India
c) To protect and preserve the Natural Environment
d) To practice Family planning and control population.
77. The Directive Principles Aim at
a) Ensuring Individual liberty
b) Ensuring strengthening of the Country's Independence
c) Providing a social and economic base for a genuine democracy in the Country.
d) Achieving all the above objectives.
78. The Directive Principles are the
a) Positive instructions to the Government to work for the attainment of the set objectives
b) Negative injunctions to the Government to refrain from encroaching on the freedom of the people.
c) Directive to the State to enhance the International prestige of the Country
d) Directives to the Government to pursue a policy of non alignment.
79. Which one of the following has been wrongly listed as Directive Principle based on 'Liberal Principles'?
a) Separation of Judiciary and Executive
b) Provision of a Uniform Civil code for the Country
c) Protection of monuments and places of artistic or Historical importance
d) None of the above has been wrongly listed.
80. The Constitution has vested the Executive power of the Union Government in
a) The President of India
b) The Prime Minister
c) The Council of Minister
d) All the above
81. Cooking Means
a) Boiling under pressure
b) Retaining results which fit theory
c) Making deceptive statements
d) Misleading the Public about quality of the product.

82. Which one is not a Trade Secret?
a) Theorem b) Equipment c) Formulae d) Pattern
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90. Engineering Ethics ,
a) Stimulates the moral imagination
b) Provides up – to – date knowledge in the field of Engineering.
c) Stimulates to Conduct Research d) Stresses on Time Management.
91. Impeachment proceedings can be initiated against the President in either House of Parliament only if a resolution signed by members of the House is moved.
a) 10 percent of total b) 25 percent of total
c) 20 percent of total d) 15 percent of total
92. Which one of the following functions of Prime – Minister has been wrongly listed?
a) He presides over the meeting of the Cabinet
b) He prepares the agenda for the meeting of the Cabinet.
c) He coordinates the working of various department
d) He chairs the meeting of the various standing and ad-hoc committees of Parliament.
93. A motion of no – confidence against the Council Ministers can be moved in the Lok – Sabha , if it is supported by atleast
a) 50 members b) 55 members
c) 100 members.
d) One – third of the total members of Lok - Sabha

- 94.** The President can call a Joint session of the two Houses of Parliament
- a) If a bill passed by one house is rejected by the other
 - b) If the amendment proposed to the bill by one house is not acceptable to the other house.
 - c) If the house does not take any action for six months on a bill remitted by the other house.
 - d) Under all the above conditions.
- 95.** The Members of the Rajya – Sabha except the nominated ones are
- a) Directly elected by the people
 - b) Elected by local Self – Governing bodies
 - c) Elected by the Legislative Assemblies of the States.
 - d) Elected partially by Legislative Assemblies and partially by the Local Self Governing bodies.
- 96.** The President who is the head of the State under the Parliamentary system prevailing in India.
- a) Enjoys absolute powers
 - b) Enjoys limited but real powers
 - c) Enjoys only nominal powers
 - d) Enjoys no powers
- 97.** The Vice – President is the Ex – Office Chairman of
- a) The Rajya Sabha
 - b) The National Development Council
 - c) The Planning Commission
 - d) None of the above
- 98.** The Speaker of the Lok – Sabha is
- a) Appointed by the President.
 - b) Appointed by the President on the recommendations of the P.M.
 - c) Elected by the members of the two houses at a joint sitting.
 - d) Elected by the members of the Lok – Sabha.
- 99.** The Rajya – Sabha is a permanent House but
- a) One – third of its members retire every two years
 - b) One – half of its members retire every three years
 - c) One – fifth of its members retire every year
 - d) One – half of its member retire every two years.
- 100.** The power to control the expenditure of the Government of India rests exclusively with
- a) The Parliament
 - b) The President
 - c) The Comptroller and Auditor General
 - d) The Union Finance Minister

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Question Paper Version : C

Third/Fourth Semester B.E. Degree Examination, Feb./Mar. 2022
Constitution of India, Professional Ethics and Cyber Law
(COMMON TO ALL BRANCHES)

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 c) Advisory Jurisdiction
 d) None of the above
38. Generally, the Governor belongs to
 a) The State where he is posted
 b) Some other State
 c) The Indian Administrative Service
 d) None of the above
39. Which of the following Legislative powers is enjoyed by the Governor of a State?
 a) He can summon or prorogue the State Legislature
 b) He can appoint one sixth of the members of the Legislative Council.
 c) He can nominate certain member of the Anglo Indian Community to the Legislative Assembly.
 d) All of above powers.
40. Engineering Ethics is a
 a) Preventive Ethics
 b) Developing
 c) Natural Ethics
 d) Scientifically developed Ethics.
41. Impeachment proceedings can be initiated against the President in either House of Parliament only if a resolution signed by members of the house is moved.
 a) 10 percent of total
 b) 25 percent of total
 c) 20 percent of total
 d) 15 percent of total
42. Which one of the following functions of Prime – Minister has been wrongly listed?
 a) He presides over the meeting of the Cabinet
 b) He prepares the agenda for the meeting of the Cabinet.
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43. A motion of no – confidence against the Council Ministers can be moved in the Lok – Sabha , if it is supported by atleast
 a) 50 members
 b) 55 members
 c) 100 members.
 d) One – third of the total members of Lok - Sabha

44. The President can call a Joint session of the two Houses of Parliament
- If a bill passed by one house is rejected by the other
 - If the amendment proposed to the bill by one house is not acceptable to the other house.
 - If the house does not take any action for six months on a bill remitted by the other house.
 - Under all the above conditions.
45. The Members of the Rajya – Sabha except the nominated ones are
- Directly elected by the people
 - Elected by local Self – Governing bodies
 - Elected by the Legislative Assemblies of the States.
 - Elected partially by Legislative Assemblies and partially by the Local Self Governing bodies.
46. The President who is the head of the State under the Parliamentary system prevailing in India.
- Enjoys absolute powers
 - Enjoys limited but real powers
 - Enjoys only nominal powers
 - Enjoys no powers
47. The Vice – President is the Ex – Office Chairman of
- The Rajya Sabha
 - The National Development Council
 - The Planning Commission
 - None of the above
48. The Speaker of the Lok – Sabha is .
- Appointed by the President.
 - Appointed by the President on the recommendations of the P.M.
 - Elected by the members of the two houses at a joint sitting.
 - Elected by the members of the Lok – Sabha.
49. The Rajya – Sabha is a permanent House but
- One – third of its members retire every two years
 - One – half of its members retire every three years
 - One – fifth of its members retire every year
 - One – half of its member retire every two years.
50. The power to control the Expenditure of the Government of India rests exclusively with
- The Parliament
 - The President
 - The Comptroller and Auditor General
 - The Union Finance Minister
51. Which one of the following does not take part in the election of the President?
- Elected members of Lok - Sabha
 - Elected members of Rajya – Sabha
 - Members of the Legislative Council
 - None of these
52. The President can be removed by impeachment procedure on the ground of violating the constitution by
- The Supreme Court
 - The Lok – Sabha only
 - Both Houses of Parliament
 - The High Court
53. The Vice – President of India is elected by the
- People
 - Members of State – Legislative Assembly
 - Members of the Rajya – Sabha
 - Members of both the Houses of Parliament at Joint sitting.

54. Who discharges the duties of the President in the event of President and Vice – President being not available?
a) The Prime Minister
b) The Chief Justice of India
c) The Speaker of Lok - Sabha
d) The Attorney General of India
55. Which one of the following can the President of India declare?
a) Emergency due to threat of War, external aggression or armed rebellion
b) Emergency due to breakdown of constitutional machinery in the State
c) Financial emergency on account of threat to the financial credit of India
d) All the above.
56. Which budget is placed first in the Parliament House
a) Railway
b) General budget
c) Financial
d) Vote of credit
57. The President can make Laws through ordinances
a) During the recess of the Parliament
b) On certain subjects even when Parliament is in session
c) Only on subjects contained in the concurrent list
d) Under no circumstances.
58. The President can grant pardon in
a) All cases of punishment by Court martial
b) All offences against laws in the Union and Concurrent list
c) All cases involving death sentence
d) All the above cases
59. If State fails to comply with the directives of the Central Government, the President can
a) Declare break – down of Constitutional machinery in the State and assume responsibility for its governance
b) Send reserve police force to secure compliance with directions
c) Dissolve the State legislature and order fresh elections
d) Can do either (a) or (b)
60. Which one of the following has been wrongly listed as Judicial power of the President of India?
a) He appoints the Chief Justice and other Judges of the Supreme Court
b) He can grant pardon, reprieve and respite to a person awarded punishment
c) He can consult the Supreme Court on any question of law or fact.
d) He can remove the Judges of Supreme – Court on ground of misconduct.
61. The Chief Election Commissioner can be removed from his office before the expiry of term by the
a) Chief Justice of India
b) Prime Minister on the recommendation of Cabinet
c) President on the recommendation of Parliament after the impeachment
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62. The quorum of minimum number of members required to hold the meetings of either Houses of Parliament is
a) One - tenth
b) One - fifth
c) One - third
d) 72 hours

- 63.** The advice of the Supreme Court is
a) Binding of the President
b) Binding on the President if it is tendered unanimously
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- 64.** The Governor reserves the Right to issue ordinances
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- 65.** The Writ of Certiorari is issued by a Superior Court
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d) To a Public authority to produce a person detained by it before the Court within 24 hours.
- 66.** Which one of the following was wrongly listed as a duty of Indian Citizens
a) To uphold and protect the Sovereign unity and integrity of the Country
b) To promote harmony and the spirit of common brotherhood among the people of India
c) To protect and pressure the natural Environment
d) To practice Family planning and control population.
- 67.** The Directive Principles Aim at
a) Ensuring Individual liberty
b) Ensuring strengthening of the Country's Independence
c) Providing a social and economic base for a genuine democracy in the Country.
- 68.** Achieving all the above objectives. The Directive Principles are the
a) Positive instructions to the Government to work for the attainment of the set objectives
b) Negative injunctions to the Government to refrain from encroaching on the freedom of the people.
c) Directive to the State to enhance the International prestige of the Country
d) Directives to the Government to pursue a policy of non alignment.
- 69.** Which one of the following has been wrongly listed as Directive Principle based on 'Liberal Principles'?
a) Separation of Judiciary and Executive
b) Provision of a Uniform Civil code for the Country
c) Protection of monuments and places of artistic or Historical importance
d) None of the above has been wrongly listed.
- 70.** The Constitution has vested the Executive power of the Union Government in
a) The President of India
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- 71.** The Fundamental Rights of a Citizen can be suspended
a) By the Parliament through a Law enacted by two – third majority
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c) By the Supreme Court
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72. Which authority can a Citizen approach for securing Right of Personal freedom
 - a) The Parliament
 - b) The President
 - c) Supreme Court alone
 - d) Both Supreme Court and High Court
73. The main objective of the Fundamental Rights is to
 - a) Ensure Independence of Judiciary
 - b) Promote a Socialist Pattern of Society
 - c) Ensure Individual liberty
 - d) Ensure all the above
74. Under which section of IT Act, stealing any digital asset or information is written a cyber crime
 - a) 65
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 - c) 67
 - d) 70
75. Fundamental duties of the Indian Citizen, were
 - a) Enshrined in the original Constitution
 - b) Added to the Constitution by the 42nd Amendment.
 - c) Added to the Constitution by the 44th Amendment.
 - d) Added to the Constitution in the wake the Supreme Court Judgment Keshavananda Bharati case with consent of all the Political parties.
76. Which one of the following Fundamental Right has been subject of maximum litigation since the inauguration of the Constitution?
 - a) Right to Freedom of Speech
 - b) Right to Constitutional Remedies
 - c) Right to Property
 - d) Right against Exploitation
77. The Fundamental Rights of Citizens were
 - a) Incorporated in the original Constitution
 - b) Outlined in an Act of Parliament in 1952
 - c) Incorporated by the 42nd Amendment
 - d) Incorporated by the 44th Amendment
78. The Fundamental Rights of Indian Citizen have been criticized on the ground that
 - a) They are hemmed in by too many restrictions.
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 - a) On the recommendation of the State Legislature
 - b) On the recommendation of the C.M.
 - c) On the recommendation of Council of Minister
 - d) If he is satisfied that the State Government cannot be carried on his accordance with the provision of the Constitution.
81. Who of the following acted as the Constitutional Advisor of the Constituent Assembly
 - a) Dr. B.R. Ambedkar
 - b) Dr. Babu Rajendra Prasad
 - c) B.N. Rao
 - d) Dr. Sachidanand Sinha
82. Which one of the following provisions of the Constitution came into force soon after its adoption on 26th November 1949?
 - a) Provision relating to Citizenship
 - b) Elections
 - c) Provisional Parliament
 - d) All the above

- 83.** The three types of Justice referred in our Preamble are
a) Social, Economic and Social b) Economic, International and Political
c) Economic, Religious and Social d) Religious, Social and Political
- 84.** What was the exact Constitutional status of the Indian Republic on January 26, 1950, when the Constitution was inaugurated?
a) A Democratic Republic b) Sovereign Democratic Republic
c) A Sovereign Secular Democratic Republic
d) A Sovereign Socialist Secular Democratic Republic.
- 85.** Right to against Exploitation seeks to protect the weaker sections of Society by
a) Giving equal pay for equal work for men and women.
b) Prohibiting human trafficking and beggar
c) Providing compulsory education for children below the age of 14 years
d) Forcing a person to work against his will without payment
- 86.** Which one of the following Directive principles can be described as Gandhian in nature?
a) Providing equal pay for equal work for both Men and Women
b) Workers participation in Management
c) Organization of Village Panchayats as units of self Government
d) Separation of Judiciary from the Executive.
- 87.** Who has been vested with the power to decide whether the restrictions imposed on the Fundamental Rights of Indian Citizen are reasonable or not
a) The Parliament b) The President
c) The Courts d) None of the above
- 88.** Which one of the following rights conferred by the Constitution is also available to Noncitizens,
a) Freedom of speech assembly and association
b) Freedom to move, reside and settle in any part of the territory of India
c) Freedom to acquire property or to carry on any occupation, trade or business
d) Right to Constitutional remedies.
- 89.** Which one of the following has been wrongly listed as a special feature of Fundamental Rights in India
a) Fundamental Rights are more sacrosanct than rights granted by ordinary laws
b) Fundamental Rights are subject to reasonable restrictions
c) Fundamental Rights are Justifiable and can be enforced through the Supreme Court
d) None of these.
- 90.** The main objective of the Cultural and Educational Rights granted to the Citizen is
a) To preserve the rich culture heritage of India.
b) To evolve a single integrated India culture.
c) To help the minorities to conserve their culture.
d) All the above.
- 91.** The Constitution of India was enacted by a Constituent Assembly set up
a) Under the Cabinet Mission Plan 1946
b) Under the Indian Independence Act 1947
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- 92.** The Members of the Constituent Assembly are
a) Directly elected by the people b) Nominated by various Political Parties
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a) Distribution of Legislative powers between the Union Government and the State Government.
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- 95.** The source of Authority of the Indian Constitution is
a) The Government of India b) The People of India
c) The President d) The Parliament
- 96.** The Preamble was amended by
a) 24th Amendment b) 42nd Amendment
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a) Spirit of brotherhood b) Fatherly treatment
c) Unity and integrity d) Elimination of Economic Justice
- 98.** In the final form of the Constitution adopted by the Constituent Assembly, how many Articles and Schedules were there?
a) 397 Articles and 7 Schedules b) 395 Articles and 4 Schedules
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a) Democratic b) Adult Franchise c) Sovereignty d) Fraternity
- 100.** The strength of the Constituent Assembly, after the withdrawal of the Muslim League, was reduced to
a) 299 members b) 329 member c) 331 members d) 359 members

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Question Paper Version : D

Third/Fourth Semester B.E. Degree Examination, Feb./Mar. 2022
Constitution of India, Professional Ethics and Cyber Law
(COMMON TO ALL BRANCHES)

Time: 2 hrs.]

[Max. Marks: 100]

INSTRUCTIONS TO THE CANDIDATES

- Answer all the hundred questions, each question carries one mark.
 - Use only **Black ball point pen** for writing / darkening the circles.
 - For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
 - Darkening two circles for the same question makes the answer invalid.
 - Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.
-
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31. Cooking Means
a) Boiling under pressure b) Retaining results which fit theory
c) Making deceptive statements
d) Misleading the Public about quality of the product.
32. Which one is not a Trade Secret?
a) Theorem b) Equipment c) Formulae d) Pattern
33. The codes of Ethics can be taken as guidelines by the Engineers to
a) Resolve the conflicts b) Formulate problems
c) Overcome the work pressure d) Escape from the responsibility
34. A Fault tree is used to
a) Assess the risk involved b) Claim compensation
c) Take free consent d) Improve safety
35. Risk of harm equal to probability of producing benefit is
a) Inevitable Risk b) Acceptable Risk
c) Risk which cannot be avoided d) None of these

36. One of the Aims of studying Engineering Ethics is to
 a) Inspire Engineers to acquire in depth knowledge in their field
 b) Acquire new skills in Engineering testing.
 c) Stimulate moral imaginations
 d) Make Engineers self – confident in discharging their duties.
37. Which one is not an impediment to responsibility?
 a) Group think
 b) Microscopic vision
 c) Trademark
 d) Egocentric tendencies
38. Being safe or blaming others is type of attitude of responsibility of Engineers
 a) Minimalist
 b) Reasonable care
 c) Good works
 d) None of these
39. To overcome an impediment ‘Uncritical Acceptance’, what step an Engineer has to take?
 a) Accept and Analyse
 b) Analyse and Accept
 c) Always say Yes Boss
 d) None of these
40. Engineering Ethics ,
 a) Stimulates the moral imagination
 b) Provides up – to – date knowledge in the field of Engineering.
 c) Stimulates to Conduct Research
 d) Stresses on Time Management.
41. For hacking a database or accessing and manipulating data which of the following language the hacker must know?
 a) SQL
 b) HTML
 c) TCL
 d) F #
42. _____ are piece of programs or scripts that allow hackers to take control over any system.
 a) Exploits
 b) Antivirus
 c) Firewall bypassers
 d) Worms
43. The process of finding vulnerabilities and exploiting them using exploitable scripts or programs are known as
 a) Infiltrating
 b) Exploitation
 c) Cracking
 d) Hacking
44. How many types of exploits are there based on their nature from hackings perspective?
 a) 04
 b) 03
 c) 02
 d) 05
45. A _____ is a set of changes done to any program or its associated data designed for updating fixing or improving it.
 a) Scratch
 b) Patch
 c) Fixer
 d) Resolver
46. Fixing of security vulnerabilities in a system by additional programs is known as _____ patches
 a) Hacking
 b) Database
 c) Server
 d) Security
47. _____ are some very frequent updates that come for every antivirus.
 a) Patch update
 b) Data update
 c) Code update
 d) Definition update
48. Cyber – Crime can be categorized into _____ types.
 a) 04
 b) 03
 c) 02
 d) 06

49. Which of the following is not a type of peer to peer cyber – crime.
a) Phishing
b) Injecting Trojans to a target victim
c) MITM
d) Credit card details leak in deep web
50. In which year India's IT Act came into existence?
a) 2000
b) 2001
c) 2002
d) 2003
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 - b) On the recommendation of the C.M.
 - c) On the recommendation of Council of Minister
 - d) If he is satisfied that the State Government cannot be carried on his accordance with the provision of the Constitution.
71. Which one of the following does not take part in the Election of the President?
 - a) Elected members of Lok - Sabha
 - b) Elected members of Rajya – Sabha
 - c) Members of the Legislative Council
 - d) None of these
72. The President can be removed by impeachment procedure on the ground of violating the Constitution by
 - a) The Supreme Court
 - b) The Lok – Sabha only
 - c) Both Houses of Parliament
 - d) The High Court

73. The Vice – President of India is elected by the
a) People
b) Members of State – Legislative Assembly
c) Members of the Rajya – Sabha
d) Members of both the Houses of Parliament at Joint sitting.
74. Who discharges the duties of the President in the event of President and Vice – President being not available?
a) The Prime Minister
b) The Chief Justice of India
c) The Speaker of Lok - Sabha
d) The Attorney General of India
75. Which one of the following can the President of India declare?
a) Emergency due to threat of War, external aggression or armed rebellion
b) Emergency due to breakdown of constitutional machinery in the State
c) Financial emergency on account of threat to the financial credit of India
d) All the above.
76. Which budget is placed first in the Parliament House
a) Railway b) General budget c) Financial d) Vote of credit
77. The President can make Laws through ordinances
a) During the recess of the Parliament
b) On certain subjects even when Parliament is in session
c) Only on subjects contained in the concurrent list
d) Under no circumstances.
78. The President can grant pardon in
a) All cases of punishment by Court martial
b) All offences against laws in the Union and Concurrent list
c) All cases involving death sentence
d) All the above cases
79. If State fails to comply with the directives of the Central Government, the President can
a) Declare break – down of Constitutional machinery in the State and assume responsibility for its governance
b) Send reserve police force to secure compliance with directions
c) Dissolve the State legislature and order fresh elections
d) Can do either (a) or (b)
80. Which one of the following has been wrongly listed as Judicial power of the President of India?
a) He appoints the Chief Justice and other Judges of the Supreme Court
b) He can grant pardon, reprieve and respite to a person awarded punishment
c) He can consult the Supreme Court on any question of law or fact.
d) He can remove the Judges of Supreme – Court on ground of misconduct.
81. Who decides disputes regarding disqualification of Members of Parliament?
a) The President
b) The Concerned house
c) The Election Commission.
d) The President in consultation with the Election Commission.

- 82.** Who presides over the Lok – Sabha if neither the Speaker nor the Deputy Speaker is not available?
- a) A member nominated by the President.
 - b) A member chosen by the Council of Minister.
 - c) A member of the panel of Chairman announced by the Speaker.
 - d) The Senior most member of the Lok – Sabha.
- 83.** Lok – Sabha is superior to the Rajya – Sabha because
- a) It is directly elected
 - b) It alone controls the Finances
 - c) It can oust the Council of Minister through a Vote of no – Confidence
 - d) of all the above reasons.
- 84.** The Supreme Court of India was setup
- a) By the Constitution
 - b) Under the Indian Independence Act 1947
 - c) Through an Act of Parliament in 1950
 - d) Under the Government of India Act 1935
- 85.** The Judges of the Supreme Court are
- a) Elected by the Parliament.
 - b) Appointed by the President on the advice of the Parliament.
 - c) Appointed by the President on the advice of the P.M.
 - d) Appointed by the President on the advice of the Chief Justice of India.
- 86.** The Judges of the Supreme Court after retirement are not permitted to carry on practice before
- a) The Supreme Court
 - b) The High Courts
 - c) The District and Session Courts
 - d) Any of the above
- 87.** Which of the following Jurisdiction of the Supreme Court of India has been wrongly listed
- a) Original Jurisdiction
 - b) Appellate Jurisdiction
 - c) Advisory Jurisdiction
 - d) None of the above
- 88.** Generally, the Governor belongs to
- a) The State where he is posted
 - b) Some other State
 - c) The Indian Administrative Service
 - d) None of the above
- 89.** Which of the following Legislative Powers is enjoyed by the Governor of a State?
- a) He can summon or prorogue the State Legislature
 - b) He can appoint one sixth of the members of the Legislative Council.
 - c) He can nominate certain member of the Anglo Indian Community to the Legislative Assembly.
 - d) All of above powers.
- 90.** Engineering Ethics is a
- a) Preventive Ethics
 - b) Developing
 - c) Natural Ethics
 - d) Scientifically developed Ethics.
- 91.** Professional Ethics is
- a) Set of Rules relating to personal character of Professionals
 - b) Traditional Rules observed since a long time.
 - c) Set of Rules passed by Professional bodies.
 - d) Set of standards adopted by Professionals.

92. Tight couple means
 - a) Binding two beams tightly
 - b) Erecting two pillars side by side
 - c) Process tightly coupled
 - d) Strong adhesive material
93. An Engineer may not be held legally liable or causing harm. When the harm is caused
 - a) Intentionally
 - b) Ignorantly
 - c) Negligently
 - d) Recklessly
94. A compound measure of the probability and magnitude of the adverse effect is known as.
 - a) Risk
 - b) Benefit
 - c) Compensation
 - d) Both (b) and (c)
95. Engineers shall issue public statements only
 - a) In subjective manner
 - b) In objective manner
 - c) On their personal responsibility
 - d) Based on the reports sent by higher Officers.
96. Attackers commonly target _____ for fetching IP address of a target or victim user.
 - a) Website
 - b) Web pages
 - c) IP tracker
 - d) Emails
97. _____ is the first phase of Ethical hacking
 - a) DNS Poisoning
 - b) Foot printing
 - c) ARP – Poisoning
 - d) Enumeration
98. Which of the following do not comes under the intangible skills of hackers?
 - a) Creative thinking
 - b) Problem solving capability
 - c) Persistence
 - d) Smart attacking potential
99. Why programming language is important for ethical hackers and security professionals?
 - a) Only to write malware.
 - b) For solving problems and building tool and programs
 - c) To teach programming
 - d) To develop program to harm others.
100. Understanding of _____ is also important for gaining access to a system through networks.
 - a) OS
 - b) Email servers
 - c) Networking
 - d) Hardware

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CBCS SCHEME

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18MATDIP31

Third Semester B.E. Degree Examination, Feb./Mar. 2022

Additional Mathematics – I

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Find the modulus and amplitude of the complex number : $\frac{(2-3i)(2+i)^2}{1+i}$. (07 Marks)
- b. Prove that $\left(\frac{1+\cos\theta+i\sin\theta}{1+\cos\theta-i\sin\theta} \right)^n = \cos n\theta + i\sin n\theta$. (06 Marks)
- c. Show that the vectors $\vec{a} - 2\vec{b} + 3\vec{c}$, $-2\vec{a} + 3\vec{b} - 4\vec{c}$, $-\vec{b} + 2\vec{c}$ are coplanar. (07 Marks)

OR

- 2 a. Given $\vec{a} = 2\hat{i} + 2\hat{j} - \hat{k}$, $\vec{b} = 6\hat{i} - 3\hat{j} + 2\hat{k}$. Find : i) $\vec{a} \cdot \vec{b}$ ii) $\vec{a} \times \vec{b}$ iii) $|\vec{a} \times \vec{b}|$. (07 Marks)
- b. Determine the value of λ , so that $\vec{a} = 2\hat{i} + \lambda\hat{j} - \hat{k}$, and $\vec{b} = 4\hat{i} - 2\hat{j} - 2\hat{k}$ are perpendicular. (06 Marks)
- c. Express $1 - i\sqrt{3}$ in the polar form and hence find its modulus and amplitude. (07 Marks)

Module-2

- 3 a. Using Euler's theorem, prove that $xu_x + yu_y = -3\cot u$ where $u = \sin^{-1} \left(\frac{x^2 y^2}{x+y} \right)$. (07 Marks)
- b. Using Maclaurin's series, prove that $\sqrt{1+\sin 2x} = 1 + x - \frac{x^2}{2} - \frac{x^3}{3} + \frac{x^4}{24} + \dots$. (06 Marks)
- c. If $u = x + 3y^2$, $v = 4x^2yz$, $w = 2z^2 - xy$, evaluate $\frac{\partial(u, v, w)}{\partial(x, y, z)}$ at the point $(1, -1, 0)$. (07 Marks)

OR

- 4 a. Obtain Maclaurin's series expansion for the function e^x upto x^4 . (07 Marks)
- b. If $u = \sin^{-1} \left(\frac{x^3 + y^3}{x+y} \right)$ prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2 \tan u$. (06 Marks)
- c. If $u = f \left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x} \right)$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$. (07 Marks)

Module-3

- 5 a. A particle moves along the curve $x = (1 - t^3)$, $y = (1 + t^2)$, $z = (2t - 5)$ determine its velocity and acceleration at $t = 1$ sec. (07 Marks)
- b. If $\vec{F} = 2x\hat{i} - 3yz\hat{j} + xz\hat{k}$, and $\phi = 2z - \frac{3}{xy}$, find $\vec{\nabla} \cdot (\vec{F} \times \vec{\nabla} \phi)$ and $\vec{\nabla} \times (\vec{F} \cdot \vec{\nabla} \phi)$ at $(1, -1, 1)$. (06 Marks)
- c. Find the constants a, b, c so that $\vec{f} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$ is irrotational. (07 Marks)

OR

- 6 a. Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ along $\vec{a} = \frac{1}{2}\hat{i} - \frac{1}{2}\hat{j} - \frac{1}{2}\hat{k}$ (07 Marks)
- b. Find $\text{curl } \vec{f}$ given that $\vec{f} = xyz^2\hat{i} + xy^2z\hat{j} + x^2yz\hat{k}$. (06 Marks)
- c. If $\vec{f} = x^2\hat{i} + y^2\hat{j} + z^2\hat{k}$ and $\vec{g} = yzi + xzj + xyk$. Show that $\vec{f} \times \vec{g}$ is a solenoidal vector. (07 Marks)

Module-4

- 7 a. Obtain the reduction formula, $I_n = \int \cos^n x dx$, where n is a positive integer. (07 Marks)
- b. Evaluate $\int_0^1 \int_x^{\sqrt{x}} xy dy dx$. (06 Marks)
- c. Evaluate $\int_0^1 \int_0^1 \int_0^1 (x + y + z) dx dy dz$. (07 Marks)

OR

- 8 a. Evaluate : $\int_0^{\pi/6} \sin^6(3x) dx$. (07 Marks)
- b. Evaluate : $\int_0^{\pi} x \sin^4 x \cos^6 x dx$. (06 Marks)
- c. Evaluate $\int_0^1 \int_0^1 \int_0^y xyz dx dy dz$. (07 Marks)

Module-5

- 9 a. Solve : $(2x + y + 1) dx + (x + 2y + 1) dy = 0$. (07 Marks)
- b. Solve : $(4xy + 3y^2 - x) dx + (x^2 + 2xy) dy = 0$. (06 Marks)
- c. Solve : $y(2xy + e^x) dx - e^x dy = 0$. (07 Marks)

OR

- 10 a. Solve : $(5x^4 + 3x^2y^2 - 2xy^3)dx + (2x^3y - 3x^2y^2 - 5y^4)dy = 0$. (07 Marks)
- b. Solve : $y(2xy + 1)dx - x dy = 0$. (06 Marks)
- c. Solve : $\frac{dy}{dx} + y \cot x = \cos x$. (07 Marks)

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MATDIP301

Third Semester B.E. Degree Examination, Feb./Mar. 2022

Advanced Mathematics – I

Time: 3 hrs.

Max. Marks:100

Note: Answer any FIVE full questions.

- 1 a. Express the complex number $\frac{2-i\sqrt{3}}{\sqrt{3}+i}$ in the form of $x + iy$. (06 Marks)
- b. Express $\sqrt{3} + i$ in the polar form and hence find modulus and amplitude. (07 Marks)
- c. Prove that $\frac{\cos\theta + i\sin\theta}{\sin\theta + i\cos\theta} = \cos 8\theta + i\sin 8\theta$. (07 Marks)
- 2 a. Find the n^{th} derivative of $e^{ax} \sin(bx + c)$. (06 Marks)
- b. If $y = \sin^{-1} x$ then prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2y_n = 0$. (07 Marks)
- c. Find the n^{th} derivative of $\frac{x + 3}{(x + 2)(2x + 3)}$. (07 Marks)
- 3 a. Find the angle between the curves $r = \sin\theta + \cos\theta$ and $r = 2\sin\theta$. (06 Marks)
- b. Find the pedal equation for the curve $r^2 = a^2 \cos 2\theta$. (07 Marks)
- c. Using Maclaurin's series expand $y = \tan x$ upto the term containing x^5 . (07 Marks)
- 4 a. If $u = \sin^{-1}\left(\frac{x^2 + y^2}{x + y}\right)$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \tan u$. (06 Marks)
- b. If $u = f(xz, y/z)$, prove that $x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} - z \frac{\partial u}{\partial z} = 0$. (07 Marks)
- c. If $x + y + z = u$, $y + z = v$ and $z = uvw$ then find the value of $\frac{\partial(x, y, z)}{\partial(u, v, w)}$. (07 Marks)
- 5 a. Obtain the reduction formula for $\int \sin^n x dx$, where n is a positive integer. (06 Marks)
- b. Evaluate $\int_0^{\pi/2} \sin^3 \theta \cos^7 \theta d\theta$. (07 Marks)
- c. Evaluate $\int_0^a \int_0^{\sqrt{a^2 - x^2}} x^2 y dy dx$. (07 Marks)
- 6 a. Evaluate $\int_0^1 \int_0^2 \int_0^3 xyz dz dy dx$. (06 Marks)
- b. Define Beta function and prove that $\beta(m, n) = \beta(n, m)$. (07 Marks)
- c. Using beta and gamma functions evaluate $\int_0^{\pi/2} \sin^4 \theta \cos^5 \theta d\theta$. (07 Marks)

- 7 a. Solve $\frac{dy}{dx} = \frac{y}{x} + \sin \frac{y}{x}$. (06 Marks)
- b. Solve $(1 + y^2)dx = (\tan^{-1} y - x)dy$. (07 Marks)
- c. Solve $(1 + e^{\frac{x}{y}})dx + e^{\frac{x}{y}}(1 - \frac{x}{y})dy = 0$. (07 Marks)
- 8 a. Solve $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = e^{2x} + 4$. (06 Marks)
- b. Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = e^{x^2}$. (07 Marks)
- c. Solve $(D^2 + D + 1)y = x^2 + x + 1$. (07 Marks)

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18CV51

ifth Semester B.E. Degree Examination, Feb./Mar. 2022 Construction Management and Entrepreneurship

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the functional areas of Management. (08 Marks)
- b. List and explain the Nature or characteristics of Planning. (08 Marks)
- c. Discuss the dependencies in a Gantt chart, with neat sketches. (04 Marks)

OR

- 2 a. With illustration, explain AOA and AON network diagrams. (08 Marks)
- b. The activity data of a project is given in the table below :

Activity	Predecessor	Duration
A	-	5
B	A	4
C	-	6
D	C	3
E	-	5
F	E	3
G	D, F	6
H	D, F	8
I	B, G	4
J	B, G	7
K	H, I	5
L	J, K	2

Draw the network diagram , Identify the critical path and Project duration using CPM. (08 Marks)

- c. Discuss the types of Management Styles. (04 Marks)

Module-2

- 3 a. Explain i) Minimum Wages Act 1948 (08 Marks) ii) The Labour Welfare Fund Act 1965 (08 Marks)
- b. Explain the importance of Resource Management in the construction of a Project. (08 Marks)
- c. Discuss the types of Maintenance. (04 Marks)

OR

- 4 a. Define Labour Production Rate or Productivity. Discuss the factors affecting Productivity. (10 Marks)
- b. The initial cost of a piece of construction equipment is Rs 35,00,000. It has a useful life of 10 years. The estimated salvage value of the equipment at the end of useful life is Rs 5,00,000. Calculate the Annual depreciation and Book value of the construction equipment using Sinking fund method. The interest rates is 8% per year. (10 Marks)

Module-3

- 5 a. Explain the process of Project Quality Management. (10 Marks)
 b. Explain the Safety measures adopted during drilling and blasting. (10 Marks)

OR

- 6 a. Discuss in detail about the cost of Quality in construction. (10 Marks)
 b. Explain the types of Conflict of Interest. (06 Marks)
 c. Discuss the following : (04 Marks)
 i) Gifts and Bribes ii) Whistle Blowing.

Module-4

- 7 a. Define Engineering Economics. Explain the principles of Engineering Economy. (08 Marks)
 b. With illustration, explain Cash Flow diagram. (06 Marks)
 c. Differentiate between Micro and Macro Economics. (06 Marks)

OR

- 8 a. Explain Break Even Analysis. Mention the assumptions of Break Even Analysis. (08 Marks)
 b. A firm has identifies two mutually exclusive investment proposals for new project whose details are given below. The life of all the two alternatives is estimated to be five years with negligible salvage value. The minimum attractive rate of return for the firm is 12%. Find the best alternative based on the rate of Return method of Comparison. (12 Marks)

	Alternative 1	Alternative 2
Investment	1,50,000/-	3,10,000/-
Annual Net Income	50,000/-	90,000/-

Module-5

- 9 a. Explain the stages in Entrepreneurial Process. (10 Marks)
 b. Discuss in detail about the Project report for starting a new Venture. (10 Marks)

OR

- 10 a. Explain the role and significance of Venture Capital. (10 Marks)
 b. Mention the objectives and functions of following agencies : (10 Marks)
 i) KIADB ii) TECSOK.

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Fifth Semester B.E. Degree Examination, Feb./Mar. 2022

Analysis of Indeterminate Structures

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Assume missing data suitably.

Module-1

- 1 Analyze the continuous beam shown in Fig.Q.1 by slope deflection method. Draw BMD and SFD. (20 Marks)

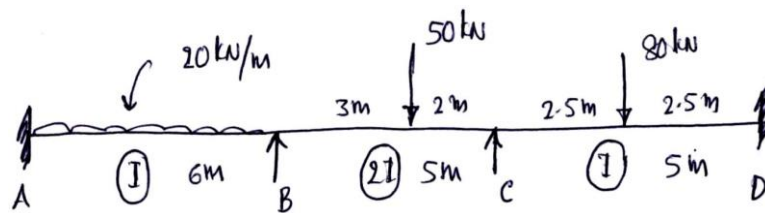


Fig.Q.1

OR

- 2 Analyze the portal frame shown in Fig.Q.2 by slope deflection method. Draw BMD. (20 Marks)

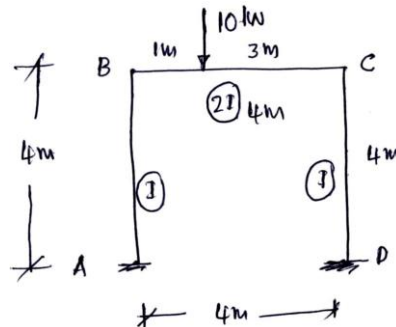


Fig.Q.2

Module-2

- 3 Analyze the beam shown in Fig.Q.3 by moment distribution method. Draw BMD EI is constant. (20 Marks)

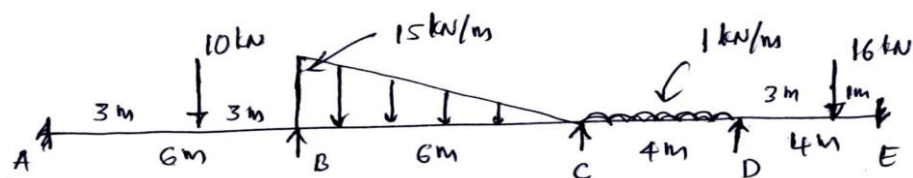


Fig.Q.3

OR

- 4 Analyze the portal frame by moment-distribution method draw BMD. (20 Marks)

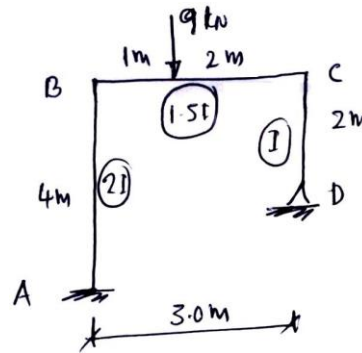


Fig.Q.4

Module-3

- 5 Analyze the continuous beam loaded shown in Fig.Q.5 by Kani's rotation method. Draw BMD. (20 Marks)

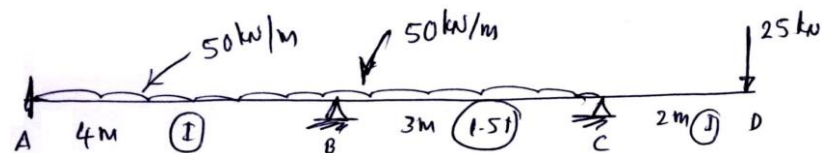


Fig.Q.5

OR

- 6 Analyze the frame shown in Fig.Q.6 by Kani's method. Take the advantage of symmetry. (20 Marks)

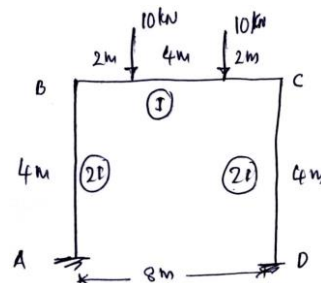


Fig.Q.6

Module-4

- 7 Analyze the continuous beam by flexibility matrix method (system approach). Draw BMD. (Fig.Q.7). (20 Marks)

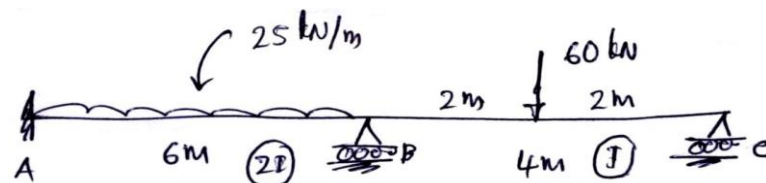


Fig.Q.7

OR

- 8 Analyze the L-frame shown in Fig.Q.8 by flexibility matrix method. Draw BMD (system approach). (20 Marks)

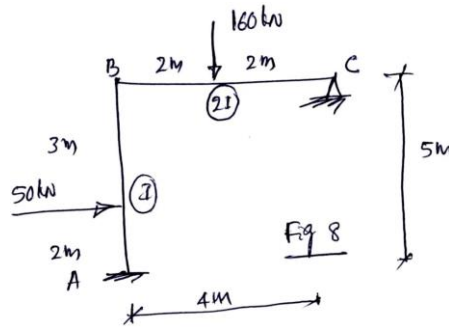


Fig.Q.8

Module-5

- 9 Analyze the continuous beam by stiffness matrix method (system approach) shown in Fig.Q.9. Draw BMD EI is constant. (20 Marks)

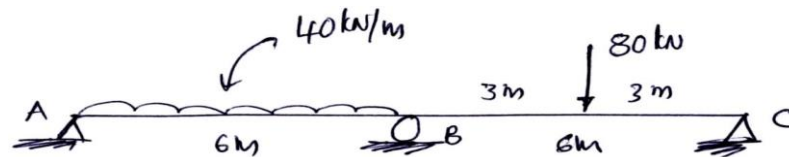


Fig.Q.9

OR

- 10 Find the forces in the members of a joint 'O' shown in Fig.Q.10 by stiffness matrix method. (system approach). (20 Marks)

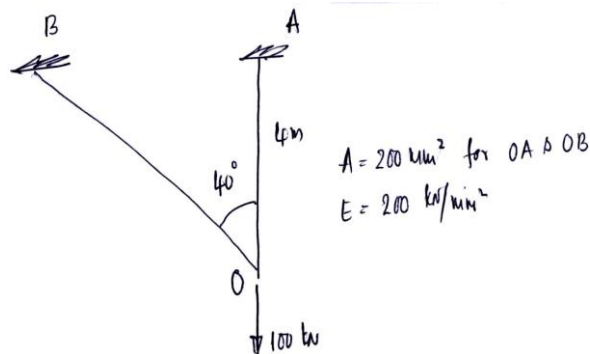


Fig.Q.10

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Fifth Semester B.E. Degree Examination, Feb./Mar. 2022 Design of RC Structural Elements

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Assume any missing data.
3. Use of IS-456, SP-16 chart permitted.

Module-1

- 1 a. Explain the following:
 - i) Partial safety factors for loads and materials.
 - ii) Explain the principles of limit state design. (10 Marks)
- b. Explain under reinforced section, over-reinforced section, balance section with neat sketches and also show that $X_{ulim} = 0.53d$, for Fe250 grade of steel. (10 Marks)

OR

- 2 a. Briefly explain the step by step procedure for short term deflection and long term deflection. (10 Marks)
- b. A flanged beam of T-section is simply supported over an effective span of 8m. The beam has effective flange width of 1400mm, thickness of flange as 150mm, breadth of web as 300mm and effective depth of 450mm. It is reinforced with 4 bars of 25mm diameter in tension and 3 bars of 16mm diameter in compression, check the beam for deflection. Use M₂₀ grade concrete and Fe415 steel. (10 Marks)

Module-2

- 3 A R.C.C beam of rectangular section 300 × 600mm is reinforced with 4 bars of 20mm diameter with an effective cover 50mm, effective span of the beam is 6m. Assuming M₂₀ concrete and Fe250 steel. Determine the central concentrated load P, that can be carried by the beam in addition to its self weight. (20 Marks)

OR

- 4 a. Find the steel for a rectangular section 300 × 600mm to support a load of 80kN/m with span of 6m (effective) and cover 40mm (effective) adopt M₂₀ concrete Fe415 steel. (10 Marks)
- b. A singly reinforced beam 250 × 500mm is reinforced with 4 – 16mm diameter and cover 40mm (effective) with effective span 6m. Determine the central point load that can be applied at mid span adopt M₂₀ concrete Fe500 steel. (10 Marks)

Module-3

- 5 A T-beam slab floor has 125mm thick slab forming part of T-beam which are of 8m clear span. The end bearing are 450mm thick, spacing of T-beams is 3.5m. The live load on the floor is 3kN/m². Design one of the intermediate beams. Use M₂₀ concrete and Fe415 steel. (20 Marks)

OR

- 6 A rectangular beam is to be simply supported on supports of 300mm width. The clear span of the beam is 6m. The beam is to have width of 230mm. The characteristic superimposed load is 12kN/m. Using M₂₀ and Fe500 steel, design the beam and sketch details of reinforcement. (20 Marks)

Module-4

- 7 A hall has clear dimensions 3m × 9m with wall thickness 230mm. The live load on the slab is 3kN/m² and finishing load 1kN/m² may be assumed. Use M₂₀ grade concrete and Fe415 steel. Design the slab, check for shear and deflection. (20 Marks)

OR

- 8 Design a dog-legged stairs for an building in a room measuring 3.6 × 5.2m clear span. The vertical distance between the floors is 3.2m. Consider LL 3kN/m². Use M20 concrete and Fe415 grade of steel. Assume stairs are supported on 300mm wall at the outer edges of landing slabs consider Rise = 160mm and Tread = 300mm. (20 Marks)

Module-5

- 9 a. Distinguish between short column and long column. (05 Marks)
b. Design a circular pin ended column 400mm diameter and helically reinforced with an unsupported length 4.5m to carry a factored load 900kN. Assume M₃₀ concrete and Fe415 steel. (15 Marks)

OR

- 10 Design a Isolated rectangular footing of uniform depth for the column size of 230mm × 300mm supporting an axial service load of 850kN. The safe bearing capacity of soil is 150kN/m². Adopt M₂₀ grade concrete and Fe415 grade steel sketch the reinforcement details. (20 Marks)

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18CV54

Fifth Semester B.E. Degree Examination, Feb./Mar. 2022 Basic Geotechnical Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the following:

(i) Water content
(iii) Unit weight of soil

(ii) Void ratio
(iv) Density index

(06 Marks)
- b. Prove the relation $\gamma_{dry} = \frac{Gr_w}{1+e}$ and $Se = wG$ from basic principles. (08 Marks)
- c. A soil has a bulk unit weight of 26 kN/m³, $G = 2.76$ and water content 16%. Determine the dry unit weight, void ratio, porosity and degree of saturation. What is the unit weight if the soil gets fully saturated due to rain? (06 Marks)

OR

- 2 a. With the help of the phase diagram, define the terms bulk density, dry density, degree of saturation and specific gravity of soil solids. (08 Marks)
- b. Following are the results obtained from the tests conducted on two soils A and B. Classify them as per IS classification system. Explain the steps involved.

Soil	W _L (%)	W _P (%)	% retained on 75 μ sieve	% retained on 4.75 mm Sieve	C _u	C _c
A	110	50	40	Zero	-	-
B	-	-	92	05	0.7	0.2

- (06 Marks)
- c. Explain the determination of specific gravity of soil solids by pycnometer method. (06 Marks)

Module-2

- 3 a. Explain different types of soil structures. (08 Marks)
- b. Discuss the factors affecting compaction in detail. (06 Marks)
- c. Determine the relative compaction of soil if the field density is 18.51 kN/m³; whose HDD is 22 kN/m³ and OMC is 13%. Comment on that. (06 Marks)

OR

- 4 a. Explain different types of clay minerals. (08 Marks)
- b. A proctor compaction test was conducted on a soil sample and the following observations were made:

W.C. (%)	8	11.5	14.5	17.5	19.5	21.5
Mass of the soil (kg)	1.70	1.90	2.0	1.98	1.95	1.92

If the volume of the mould is 950 C.C. and specific gravity of soil was 2.65, draw the:

- (i) Dry density v/s moisture content curve and get the maximum dry density and optimum moisture content.
- (ii) 100% saturation line.
- (iii) Also calculate the minimum void ratio and saturation at OMC. (12 Marks)

Module-3

- 5 a. List the factors affecting permeability in soils. Explain any four. (08 Marks)
- b. An earth dam is built on an impervious foundation with a horizontal filter under the downstream slope. The horizontal and vertical permeabilities of the soil material in the dam are 4×10^{-3} and 1×10^{-3} cm/s respectively. The full reservoir level is 15 m above downstream filter. A flow net, constructed for the transformed section of the dam, consists of 4 flow channels and 15 equipotential drops. Estimate the seepage loss per m length of the dam. (08 Marks)
- c. A stratum of fine sand is 2m thick. Under what head of water, flowing in an upward direction will the quick condition develop? Take $G = 2.68$ and $e = 0.6$. (04 Marks)

OR

- 6 a. Explain Casagrande's method of establishing the phreatic line of an earth dam with horizontal drainage filter on the downstream side. (08 Marks)
- b. Derive the expression for the determination of coefficient of permeability of the soil by falling head method. (06 Marks)
- c. Differentiate between: (i) Total stress (ii) Pore water pressure (iii) Effective stress (06 Marks)

Module-4

- 7 a. Derive the relation $\sigma_1 = \sigma_3 \tan^2 \alpha + 2c \tan \alpha$. (08 Marks)
- b. A shear box test conducted on a soil sample gives the following observations:

Normal load (N)	360	720	1080	1440
Shear load proving ring dial reading (divisions)	13	19	26	32

If the shear box is 60 mm square and proving ring constant is 20 N per division. Find out the shear parameters (C & ϕ) of the soil in kN/m^2 and degrees respectively. (12 Marks)

OR

- 8 a. Write a note on Vane Shear test. (08 Marks)
- b. The following data relate to a triaxial compressive test performed on a soil samples.

Test No.	Confining Pressure (kN/m^2)	Deviatric Stress (kN/m^2)
1	80	175
2	150	240
3	210	300

Determine the total stress parameters of the soil. (12 Marks)

Module-5

- 9 a. What is pre-consolidation pressure? How it is determined by Casagrande's method? (08 Marks)
- b. Explain pre-consolidated, normally consolidated and under consolidated soil. (04 Marks)
- c. A 30 cm thick sample of clay reached 30% consolidation in 15 minutes with drainage both at top and bottom. How long will it take the clay layer from which the sample was obtained to reach 50% consolidation? The clay layer has one-way drainage and was 6 m thick. (08 Marks)

OR

- 10 a. Explain the determination of coefficient of consolidation by square root of time fitting method. (08 Marks)
- b. A 3 m thick layer of clay was subjected to a loading of 0.7 kg/cm^2 . It attained 50% consolidation after 1 year. The layer had double drainage. Determine:
 (i) Coefficient of consolidation (ii) Settlement after one year if $k = 5 \text{ mm/yr}$
 (iii) Time required for 90% consolidation (12 Marks)

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CBCS SCHEME

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18CV55

Fifth Semester B.E. Degree Examination, Feb./Mar. 2022 Municipal Wastewater Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the different types of sewerage systems with their advantages, disadvantages and suitability. (10 Marks)
b. Explain Dry Weather Flow (DWF) and explain the factors on which DWF depends. (10 Marks)

OR

- 2 a. Define sewer appurtenances. Explain with neat sketch, construction and working of manhole. (10 Marks)
b. Explain any five different shapes of sewers with neat sketches. (10 Marks)

Module-2

- 3 a. Design a sewer for a population of 50,000 with per capita water supply of 150 lit/day. The slope available is 1 in 500 and 80% of the water supplied emerges as sewage. The sewer is to be designed to carry 4 times the DWF when running full. Assume $N = 0.012$ and compute the velocity of flow when running full. (10 Marks)
b. Explain self cleaning velocity and non-scouring velocity. What are different types of sampling? Explain. (10 Marks)

OR

- 4 a. Draw a neat flow diagram employed for a municipal wastewater treatment plant. Indicate the importance of each unit indicated in the flow diagram. (10 Marks)
b. The 5 day 30°C BOD of sewage sample is 110 mg/l. Calculate its 5 days 20°C BOD. Assume the deoxygenation constant at 20°C , K_{20} as 0.1. (06 Marks)
c. Explain the term BOD and their importance in wastewater treatment. (04 Marks)

Module-3

- 5 a. Explain the working of a "Grit Chamber" and "Oil and Grease" removal tank with figures. (10 Marks)
b. Write short notes on: (i) Screens (ii) Settling tank (10 Marks)

OR

- 6 a. Discuss in detail the process of de-oxygenation and re-oxygenation with respect to self-purification of natural water with a neat sketch. (10 Marks)
b. A city discharges 100 cumecs of sewage into a river which is fully saturated with oxygen and flowing at the rate of 1500 cumecs during its lean days with a velocity of 0.1 m/sec. The 5-days BOD of sewage at the given temperature is 280 mg/l. Find when and where the critical D.O. deficit will occur in the downstream portion of the river, and what is its amount. Assume coefficient of purification of the stream (f) as 4.0 and coefficient of de-oxygenation (K_D) as 0.1. (Take saturated D.O. = 9.2 mg/l). (10 Marks)

Module-4

- 7 a. Explain the five modifications of activated sludge processes. (10 Marks)
- b. The sewage is flowing at 4.5 million litres per day from a primary clarifier to a standard rate trickling filter. The 5-day BOD of the influent is 160 mg/l. The value of the adopted organic loading is to 160 gm/m³/day, and surface loading 2000 l/m²/day. Determine the volume of the filter and its depth. Also calculate the efficiency of this filter unit. (10 Marks)

OR

- 8 a. Explain briefly with neat sketches, the working of :
 (i) Sludge digester's
 (ii) Sludge drying beds (10 Marks)
- b. Calculate the dimensions of an oxidation pond for treating sewage from a residential colony with a population of 5000 persons. Assume the rate of sewage flow 120 lpcd and 5 day BOD of sewage as 300 mg/l. Take organic loading as 300 kg/ha/day and length of the tank as twice of its width and depth of pond as 1.2 m. Apply check for detention time. (10 Marks)

Module-5

- 9 a. Write about the need for advanced wastewater treatment? Explain the biological phosphorus removal process. (10 Marks)
- b. What is the necessity for the removal of nitrogen? Discuss the nitrification and denitrification process for removal of nitrogen. (10 Marks)

OR

- 10 a. Explain the septic tank with neat sketch. Also write the design considerations required for septic tank. (10 Marks)
- b. Write brief note on with sketch:
 (i) Two-pit latrines
 (ii) Soak pits (10 Marks)

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GBCS SCHEME

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18CV56

Fifth Semester B.E. Degree Examination, Feb./Mar.2022

Highway Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What are the characteristics of road transport in comparison with other systems? (06 Marks)
- b. List the significant recommendations of Jayakar committee. What are the implementations based on Jayakar committee recommendation. (06 Marks)
- c. For the following data of population units of 0.5, 1.0, 2.0 and 1.0 per 1000 tonnes, 500 tonnes and 100 tonnes of agricultural, raw material and industrial products respectively. Find the priority for the following system of roads.

Proposal	Length (km)	Population range			Productivity		
		<1000	1001 – 2000	> 2000	Agri	Raw material	Industrial Products
A	25	20	15	25	8000	4000	1000
B	35	30	20	40	6000	1000	1600
C	40	50	20	60	4500	2000	3200
D	30	15	12	30	4000	6000	500

(08 Marks)

OR

- 2 a. What is an ideal alignment? Explain with neat sketches, how you will align through, (i) Hill pass, (ii) Bridge site (iii) Marshy land. (08 Marks)
- b. Determine the lengths of different categories of roads in a state in India by the year 2001, using the following data:
Area of state : 15000 km²
No. of towns : 20 numbers
Road density : 82 km/ 100 km². (08 Marks)
- c. List the salient features of KSHIP, PMGSY (04 Marks)

Module-2

- 3 a. Sketch a neat cross section of NH in rural section in embankment and cutting. (06 Marks)
- b. Calculate the minimum sight distance required to avoid a head on collision of two cars approaching from opposite directions at 90 kmph and 60 kmph. Assume the reaction time of 2.5 sec, co-efficiency of friction of 0.65 and a break efficiency of 55% in either case. (08 Marks)
- c. List the factors affecting skid resistance. (06 Marks)

OR

- 4 a. The speed of overtaking and overtaken vehicles are 80 and 50 kmph respectively on a 2-way traffic road. If acceleration of overtaking vehicle is 0.99 m/sec². Calculate OSD, mention minimum length of overtaking zone, draw a neat sketch of the minimum overtaking zone and show the positions of the sign post. (10 Marks)
- b. Derive an expression for super elevation such that speed can be substituted in kmph. (05 Marks)
- c. An ascending gradient of 1 in 100 meets a descending gradient of 1 in 120. A Summit curve is to be designed for a speed of 80 kmph so as to have an overtaking sight distance of 470 m. (05 Marks)

Module-3

- 5 a. List and briefly explain the desirable properties of subgrade soil. (05 Marks)
 b. List and explain desirable properties of Aggregates to be used in road construction. (05 Marks)
 c. CBR tests were conducted on two specimens of a soil. Determine CBR value of the soil if 100 division of the load dia/proving ring represents 190 kg load in the calibration chart of the proving ring. Use following data:

Penetration of plunger (mm)	Load dial readings, divisions	
	Specimen - 1	Specimen - 2
	0	0
0.5	8	0.5
1.0	15	1.5
1.5	23	2.5
2.0	29	6.0
2.5	34	13
3.0	37	20
4	43	30
5	48	38
7.5	57	50
10	63	58
12.5	67	63

(10 Marks)

OR

- 6 a. Differentiate between Tar and Bitumen. (05 Marks)
 b. List the basic structural components of a flexible pavement and briefly explain the function of the same with a neat figure. (08 Marks)
 c. Calculate ESWL of a dual wheel assembly carrying 2004 kg each for pavement thickness of 15, 20 cm. Centre to centre spacing = 27 cm and distance between the walls of tyres = 11 cm. (07 Marks)

Module-4

- 7 a. Briefly explain Ruthfutch method. (10 Marks)
 b. Describe step by step procedure for subgrade preparation, with proper specifications. (10 Marks)

OR

- 8 a. With detailed procedure, describe the method of constructing WBM. (08 Marks)
 b. Write the specification for Bituminous concrete also provide step by step procedure for laying Bituminous concrete layer. (09 Marks)
 c. Define Tack coat, Seal coat, Prime coat. (03 Marks)

Module-5

- 9 a. Briefly explain the significance of highway drainage. (10 Marks)
 b. With neat sketches, explain lowering of water table, control of seepage flow.[Clay seal method and transverse method] (10 Marks)

OR

- 10 a. With a neat relevant sketch, explain the procedure for design of filter material in Highway drainage. (10 Marks)
 b. Explain briefly the quantifiable benefits for a highway user. (06 Marks)
 c. Write short notes on BOT and BOOT concepts. (04 Marks)

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Question Paper Version : A

Fifth Semester B.E. Degree Examination, Feb./Mar. 2022

Environmental Studies**(COMMON TO ALL BRANCHES)**

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the hundred questions, each question carries **ONE mark**.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. The word Environment is derived from
a) Greek b) French c) Spanish d) English
 2. Primary consumer
a) Herbivores b) Carnivores c) Macro consumer d) Omnivores
 3. Abiotic component includes
a) Soil b) Temperature c) Water d) All the three
 4. Atmosphere consists of 79% Nitrogen and 21% oxygen by
a) Volume b) Weight c) Density d) All the three
 5. Which of the following is a biotic component of an eco system
a) Fungi b) Solar light c) Temperature d) Humidity
 6. Which pyramid is always upright
a) Energy b) Biomass c) Number d) Food chain
 7. Which of the following conceptual sphere of environment is having the least storage capacity for matter.
a) Atmosphere b) Lithosphere c) Hydrosphere d) Biosphere

8. The objective of environment education is
 - a) Raise consciousness about environmental conditions
 - b) To teach environmentally appropriate behavior
 - c) Create an environmental ethic.
 - d) All of the above
9. The sequence of eating and being eaten in an ecosystem is called
 - a) Food chain
 - b) Carbon cycle
 - c) Hydrological cycle
 - d) Anthroposystem
10. The largest reservoir of Nitrogen in our planet is
 - a) Ocean
 - b) Atmosphere
 - c) Biosphere
 - d) Hydrosphere
11. Which of the following is a producer in an ecosystem
 - a) Plant and some bacteria capable of producing their own food
 - b) Animals
 - c) Humanbeing
 - d) Fish
12. The basic requirements of human being are provided by
 - a) Industrialization
 - b) Agriculture
 - c) Nature
 - d) Urbanization
13. In an ecosystem biological cycling of materials is maintained by
 - a) Producer
 - b) Consumer
 - c) Decomposer
 - d) All of these
14. A predator is
 - a) An animal that is fed upon
 - b) An animal that feeds producers
 - c) An animal that hunts and feeds on their animals
 - d) A primary consumer
15. The major atmospheric gas layer in stratosphere is
 - a) Hydrogen
 - b) Carbon-dioxide
 - c) Ozone
 - d) Helium
16. Which atmospheric sphere is closest to the earth surface
 - a) Troposphere
 - b) Starto sphere
 - c) Mesosphere
 - d) Exosphere
17. World environmental day is on
 - a) 5th May
 - b) 5th June
 - c) 22nd July
 - d) 5th July
18. Which of the following is absorbed by green plants from the atmosphere
 - a) Carbon dioxide
 - b) Water
 - c) Nutrients
 - d) All of the above
19. Habitat refer to
 - a) Physical conditions of the place where organism live
 - b) Chemical conditions of the place where organism live.
 - c) Both a and b
 - d) None of a and b
20. A food web consists of
 - a) A portion of a food chain
 - b) An organisms position in a food chain
 - c) Interlocking of food chain
 - d) A set of similar consumer

21. In an ecosystem the flow of energy is
 a) Bidirectional b) Cyclic c) Unidirectional d) Multidirectional
22. Estuary means
 a) Rich in nutrients
 b) End point of the river
 c) Meeting place of river and sea
 d) Treatment of water
23. What percentage of its geographical area of a country should be under forest cover?
 a) 23% b) 43% c) 13% d) 33%
24. Mineral resources are
 a) Renewable b) Available in plenty c) Non-renewable d) Equally distributed
25. Which of the following is considered as an alternate fuel
 a) CNG b) Kerosene c) Coal d) Petrol
26. Chernobyl nuclear disaster occurred in the year
 a) 1984 b) 1972 c) 1986 d) 1987
27. Electromagnetic radiation can cause
 a) Plague b) Malaria c) Cancer d) Fever
28. OTEC is an energy technology that convert
 a) Energy in large tides ocean to generate electricity
 b) Energy in ocean wave to generate electricity
 c) Energy in ocean due to thermal gradient to generate electricity
 d) Energy in the fast moving ocean current to generate
29. Direct conversion of solar energy is attained by
 a) Solar photo voltaic system
 b) Solar hybrid system
 c) Solar thermal system
 d) Solar air heater
30. Hydrogen energy can be trapped through
 a) Heat pump b) Fuel cell
 c) Photovoltaic cell d) Cell
31. Cow dung can be used
 a) As manure b) For production of bio gas
 c) As fuel d) All of the above
32. Natural gas contains
 a) CO₂ b) H₂ c) CH₄ d) N₂
33. Nuclear waste is active for
 a) 10 years b) 50 years c) 70 years d) Centuries
34. Molasses from sugar industry is used to generate
 a) Biodiesel b) Hydrogen c) Bio-ethanol d) Bio methanol
35. Nuclear power plant in Karnataka is located at
 a) Bhadravathi b) Kaiga c) Raichur d) KGF

36. Which resources are in exhaustible
a) Renewable b) Fossil fuel c) Non Renewable d) Mineral
37. Earth day is held every year on
a) April 22nd b) 5th June c) November 23rd d) January 26th
38. Oil and gas are preferred because of
a) Easy transportation b) Cheap c) Strong smell d) All of these
39. Mining means
a) To conserve and preserve minerals
b) To check pollution due to mineral resources
c) To extract minerals and ores
d) None of these
40. What would you do to prevent environmental damage
a) Plant trees
b) Halt deforestation
c) Control pollution
d) All of the above
41. Effect of modern agriculture on soil is due to
a) Erosion b) Acidification c) Salinization d) All
42. Bio – remediation means removal of containments from
a) Soil b) Wastewater c) Ground water d) All
43. Population explosion will cause
a) Bio diversity
b) Stress in ecosystem
c) Unemployment
d) None of these
44. Smog is combination of
a) Smoke and Snow b) Snow and Fog
c) Smoke and Fog d) All the above
45. Physical pollution of water is due to
a) pH b) Chloride c) Turbidity d) All of these
46. Renewable energy is
a) Primary source b) Secondary source
c) Tertiary source d) none of the above
47. Generation of wind energy is mainly based on
a) Storm b) Pressure
c) Velocity of wind d) Direction of wind
48. Which of the following is not a green house gas?
a) Ozone b) Water vapour c) CO₂ d) Sulphur dioxide
49. World ozone day is being celebrated on every year
a) Sept 5th b) Sept 16th c) Oct 15th d) June 5th

50. First International earth summit was held at
a) Germany b) Stock Holm c) Riode Janeiro d) England
51. Environmental protection is a fundamental duty of the citizen of India under the
a) 21 b) 47 c) 48-A d) Article 51-A(9)
52. Important sources of land pollution are
a) Industrial waste b) Agricultural waste
c) Both (a) and (b) d) None of the above
53. The radiations absorbed by ozone layer are
a) Ultra violet b) Infra red c) Gamma rays d) Visible
54. The first major environmental Protection Law promulgated in India was
a) Air Act b) Water Act
c) Environmental Act d) None of these
55. Which of the following is the authority to monitor state industrial effluents?
a) State pollution control board
b) Indian environmental association
c) Centre for science and development
d) None of these
56. What is the permissible range of pH for drinking water as per Indian standards
a) 6 to 9 b) 6.5 to 7.5 c) 6 to 8 d) 6.5 to 8.5
57. What is the maximum allowable concentration of fluorides in drinking water
a) 1.0 mg/l b) 1.25 mg/l c) 1.5 mg/l d) 1.75 mg/l
58. Which of the following is a non point source of water pollution
a) Factories b) Sewage treatment plants
c) urban and suburban lands d) All of the above
59. Nitrogen fixing bacteria exists in – of plants
a) Leaf b) Roots c) Stem d) Flower
60. Forest rich area in Karnataka is found in _____
a) Western Ghats b) Kodagu c) Mangalore d) Belagavi
61. Cholera and typhoid are caused by
a) Worms b) Virus c) Bacteria d) Fungus
62. Hepatitis is caused by
a) Protozoa b) Virus c) Fungus d) Bacteria
63. The required iron content in drinking water as specified by BIS is
a) 300 mg/l b) 30 mg/l c) 3 mg/l d) 0.3 mg/l
64. The pH value of the acid rain water is less than
a) 3.2 b) 5.7 c) 7.0 d) 8.7
65. Reduction of brightness of famous Taj Mahal is due to
a) Global warming b) Air pollution c) Ozone depletion d) Afforestation

- 66 Ozone layer thickness is measured in
a) PPM b) PPB c) Decibels d) Dobson unit
- 67 Ferons are
a) HFC b) CFC c) NFC d) Hydrocarbon
- 68 Bhopal gas tragedy caused due to leakage of
a) Methyl iso cyanate (MIC)
b) Sulphur dioxide
c) Mustard gas
d) Chlorine gas
- 69 ICDS is a welfare scheme for
a) Public b) Women c) Men d) Children
- 70 Study of trends in human population growth and prediction of future growth is called
a) Demography b) Biography c) Kalography d) Psychology
- 71 Biochemical oxygen demand measures
a) Industrial pollution
b) Air pollution
c) Polluting capacity of effluents
d) Dissolved O₂ needed by microbes to decompose organic wastes
- 72 Increasing skin cancer and high mutation rate are the result of
a) Ozone depletion b) Acid rain c) CO₂ pollution d) CO pollution
- 73 Sound become hazardous noise pollution at decibels
a) Above 30 b) Above 80 c) Above 100 d) Above 120
- 74 The zero population growth due to equal birth and death rates is called
a) Natural increase b) Demographic transition
c) Fertility rate d) Replacement level
- 75 Cauvery water dispute is in between
a) Karnataka and Andra Pradesh b) Karnataka and Maharashtra
c) Karnataka and Tamil Nadu d) None of these
- 76 Khetri (Rajasthan) is famous for
a) Gold mine b) Copper mine c) Granite stone d) Marble stone
- 77 The Water (Prevention and control of pollution) Act was enacted in the year
a) 1986 b) 1974 c) 1994 d) 1984
- 78 The Forest (conservation) Act was enacted in the year
a) 1986 b) 1974 c) 1980 d) 1972
- 79 Environmental protection is the responsibility of
a) Govt. of India b) NGOs c) Individual d) All
- 80 Which of the following is NGO
a) Narmada Bachao Andolan b) Pollution control board
c) Gove. Conservation board d) None of these

- 81 The leader of Chipko movement is
a) Sunderlal Bahuguna b) Medha Patkar c) Vandana Shiva d) Anna Hazare
- 82 The Tiger conservation project was started in
a) 1973 b) 1984 c) 1999 d) 2004
- 83 An international conference on environmental education was held in December 1982 at
a) Kyoto b) Vienna c) New Delhi d) Rome
- 84 ISO 14000 standards deals with
a) Population management b) Risk management
c) Environmental management d) None of the above
- 85 The First International Earth Summit was held at
a) Johannesburg b) Rio de Janeiro c) Kyoto d) Stockholm
- 86 Disease caused by eating fish inhabiting mercury contaminated water is
a) Bright's disease b) Minamata disease
c) Hashimoto disease d) Osteosclerosis
- 87 Nitrogen oxide and hydrocarbons released by automobiles interact to form
a) Carbon monoxide b) Ammonia c) PAN d) Aerosols
- 88 The world AIDS day is recalled on
a) 1st July b) 5th June c) 1st December d) 5th October
- 89 ENVIS Means
a) Environmental Information System
b) Environmental Interference service
c) Environmental Information Service
d) None of the above
- 90 Corona viruses is a
a) DNA viruses b) RNA viruses
c) Chromosomal viruses d) None of the above
- 91 Where did the first case of corona virus diseases originate
a) Yamen city b) Wuhan city c) Whattman city d) Wang city
- 92 SARS refer to
a) Severe acute respiratory syndrome
b) Self acute respiratory system
c) Severe acute respiratory system
d) Self accurate rest syndrome
- 93 Effect of carbon monoxide on blood, causing
a) H_2CO_3 b) COHb c) CO_2Hb d) HbCO_2
- 94 Which of the following is the most stable ecosystem
a) Mountain b) Desert c) Forest d) Ocean
- 95 In aquatic ecosystem phytoplankton can be considered as a
a) Consumer b) Producer
c) Saprotrophic organisms d) Macro consumer

- 96 A primary succession on bare rock starts with
a) Lichens b) Herbs c) Trees d) Animals
- 97 Which one is matching sanctuary
a) Kaziranga – musk deer b) Gir -Lion
c) N.E. Himalayan region – Samber d) Sunderban – rhino
- 98 Which group of vertebrates comprises the highest number of endangered species
a) Birds b) Mammals c) Fishes d) Reptiles
- 99 In our country the percentage of land under forest is about
a) 20% b) 19% c) 25% d) 30%
- 100 Fluoride pollution mainly affects
a) Kidney b) Brain c) Heart d) Teeth

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Question Paper Version : B

Fifth Semester B.E. Degree Examination, Feb./Mar. 2022

Environmental Studies**(COMMON TO ALL BRANCHES)**

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the hundred questions, each question carries **ONE mark**.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Where did the first case of corona virus dieses originate
a) Yamen city b) Wuhan city c) Whattman city d) Wang city
 2. SARS refer to
a) Severe acute respiratory syndrome
b) Self acute respiratory system
c) Severe acute respiratory system
d) Self accurate rest syndrome
 3. Effect of carbon monoxide on blood, causing
a) H_2CO_3 b) COHb c) CO_2Hb d) $HbCO_2$
 4. Which of the following is the most stable ecosystem
a) Mountain b) Desert c) Forest d) Ocean
 5. In aquatic ecosystem phytoplankton can be considered as a
a) Consumer b) Producer
c) Saprotropic organisms d) Macro consumer
 6. A primary succession on bare rock starts with
a) Lichens b) Herbs c) Trees d) Animals
 7. Which one is matching sanctuary
a) Kaziranga – musk deer b) Gir -Lion
c) N.E. Himalayan region – Samber d) Sunderban – rhino

8. Which group of vertebrates comprises the highest number of endangered species
a) Birds b) Mammals c) Fishes d) Reptiles
9. In our country the percentage of land under forest is about
a) 20% b) 19% c) 25% d) 30%
10. Fluoride pollution mainly affects
a) Kidney b) Brain c) Heart d) Teeth
11. The leader of Chipko movement is
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 a) Soil b) Wastewater c) Ground water d) All

- 53 Population explosion will cause
 a) Bio diversity
 b) Stress in ecosystem
 c) Unemployment
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- 54 Smog is combination of
 a) Smoke and Snow
 b) Snow and Fog
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 a) Primary source
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 b) 5th June
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b) Halt deforestation
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c) Energy in ocean due to thermal gradient to generate electricity
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- 84** A predator is
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- 85** The major atmospheric gas layer in stratosphere is
 a) Hydrogen b) Carbon-dioxide c) Ozone d) Helium
- 86** Which atmospheric sphere is closest to the earth surface
 a) Troposphere b) Starto sphere c) Mesosphere d) Exosphere
- 87** World environmental day is on
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- 88** Which of the following is absorbed by green plants from the atmosphere
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 a) Physical conditions of the place where organism live
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- 90** A food web consists of
 a) A portion of a food chain
 b) An organisms position in a food chain
 c) Interlocking of food chain
 d) A set of similar consumer
- 91** The word Environment is derived from
 a) Greek b) French c) Spanish d) English
- 92** Primary consumer
 a) Herbivores b) Carnivores c) Macro consumer d) Omnivores
- 93** Abiotic component includes
 a) Soil b) Temperature c) Water d) All the three
- 94** Atmosphere consists of 79% Nitrogen and 21% oxygen by
 a) Volume b) Weight c) Density d) All the three

- 95** Which of the following is a biotic component of an eco system
a) Fungi b) Solar light c) Temperature d) Humidity
- 96** Which pyramid is always upright
a) Energy b) Biomass c) Number d) Food chain
- 97** Which of the following conceptual sphere of environment is having the least storage capacity for matter.
a) Atmosphere b) Lithosphere c) Hydrosphere d) Biosphere
- 98** The objective of environment education is
a) Raise consciousness about environmental conditions
b) To teach environmentally appropriate behavior
c) Create an environmental ethic.
d) All of the above
- 99** The sequence of eating and being eaten in an ecosystem is called
a) Food chain b) Carbon cycle
c) Hydrological cycle d) Anthroposystem
- 100** The largest reservoir of Nitrogen in our planet is
a) Ocean b) Atmosphere c) Biosphere d) Hydrosphere

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Question Paper Version : C

Fifth Semester B.E. Degree Examination, Feb./Mar. 2022

Environmental Studies**(COMMON TO ALL BRANCHES)**

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the hundred questions, each question carries **ONE mark**.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

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- 61 Where did the first case of corona virus dieses originate
 a) Yamen city b) Wuhan city c) Whattman city d) Wang city
- 62 SARS refer to
 a) Severe acute respiratory syndrome
 b) Self acute respiratory system
 c) Severe acute respiratory system
 d) Self accurate rest syndrome
- 63 Effect of carbon monoxide on blood, causing
 a) H_2CO_3 b) COHb c) CO_2Hb d) HbCO_2
- 64 Which of the following is the most stable ecosystem
 a) Mountain b) Desert c) Forest d) Ocean
- 65 In aquatic ecosystem phytoplankton can be considered as a
 a) Consumer b) Producer
 c) Saprotropic organisms d) Macro consumer
- 66 A primary succession on bare rock starts with
 a) Lichens b) Herbs c) Trees d) Animals
- 67 Which one is matching sanctuary
 a) Kaziranga – musk deer b) Gir -Lion
 c) N.E. Himalayan region – Samber d) Sunderban – rhino

- 68 Which group of vertebrates comprises the highest number of endangered species
a) Birds b) Mammals c) Fishes d) Reptiles
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Question Paper Version : D

Fifth Semester B.E. Degree Examination, Feb./Mar. 2022

Environmental Studies**(COMMON TO ALL BRANCHES)**

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

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a) Troposphere	b) Starto sphere	c) Mesosphere	d) Exosphere
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a) Plant trees b) Halt deforestation c) Control pollution d) All of the above
21. Environmental protection is a fundamental duty of the citizen of India under the
a) 21 b) 47 c) 48-A d) Article 51-A(9)

22. Important sources of land pollution are
 - a) Industrial waste
 - b) Agricultural waste
 - c) Both (a) and (b)
 - d) None of the above
23. The radiations absorbed by ozone layer are
 - a) Ultra violet
 - b) Infra red
 - c) Gamma rays
 - d) Visible
24. The first major environmental Protection Law promulgated in India was
 - a) Air Act
 - b) Water Act
 - c) Environmental Act
 - d) None of these
25. Which of the following is the authority to monitor state industrial effluents?
 - a) State pollution control board
 - b) Indian environmental association
 - c) Centre for science and development
 - d) None of these
26. What is the permissible range of pH for drinking water as per Indian standards
 - a) 6 to 9
 - b) 6.5 to 7.5
 - c) 6 to 8
 - d) 6.5 to 8.5
27. What is the maximum allowable concentration of fluorides in drinking water
 - a) 1.0 mg/l
 - b) 1.25 mg/l
 - c) 1.5 mg/l
 - d) 1.75 mg/l
28. Which of the following is a non point source of water pollution
 - a) Factories
 - b) Sewage treatment plants
 - c) urban and suburban lands
 - d) All of the above
29. Nitrogen fixing bacteria exists in – of plants
 - a) Leaf
 - b) Roots
 - c) Stem
 - d) Flower
30. Forest rich area in Karnataka is found in _____
 - a) Western Ghats
 - b) Kodagu
 - c) Mangalore
 - d) Belagavi
31. Biochemical oxygen demand measures
 - a) Industrial pollution
 - b) Air pollution
 - c) Polluting capacity of effluents
 - d) Dissolved O₂ needed by microbes to decompose organic wastes
32. Increasing skin cancer and high mutation rate are the result of
 - a) Ozone depletion
 - b) Acid rain
 - c) CO₂ pollution
 - d) CO pollution
33. Sound become hazardous noise pollution at decibels
 - a) Above 30
 - b) Above 80
 - c) Above 100
 - d) Above 120
34. The zero population growth due to equal birth and death rates is called
 - a) Natural increase
 - b) Demographic transition
 - c) Fertility rate
 - d) Replacement level
35. Cauvery water dispute is in between
 - a) Karnataka and Andhra Pradesh
 - b) Karnataka and Maharashtra
 - c) Karnataka and Tamil Nadu
 - d) None of these

36. Khetri (Rajasthan) is famous for
a) Gold mine b) Copper mine c) Granite stone d) Marble stone
37. The Water (Prevention and control of pollution) Act was enacted in the year
a) 1986 b) 1974 c) 1994 d) 1984
38. The Forest (conservation) Act was enacted in the year
a) 1986 b) 1974 c) 1980 d) 1972
39. Environmental protection is the responsibility of
a) Govt. of India b) NGOs c) Individual d) All
40. Which of the following is NGO
a) Narmada Bachao Andolan
b) Pollution control board
c) Gove. Conservation board
d) None of these
41. Where did the first case of corona virus dieses originate
a) Yamen city b) Wuhan city c) Whattman city d) Wang city
42. SARS refer to
a) Severe acute respiratory syndrome
b) Self acute respiratory system
c) Severe acute respiratory system
d) Self accurate rest syndrome
43. Effect of carbon monoxide on blood, causing
a) H_2CO_3 b) COHb c) CO_2Hb d) $HbCO_2$
44. Which of the following is the most stable ecosystem
a) Mountain b) Desert c) Forest d) Ocean
45. In aquatic ecosystem phytoplankton can be considered as a
a) Consumer b) Producer
c) Saprotropic organisms d) Macro consumer
46. A primary succession on bare rock starts with
a) Lichens b) Herbs c) Trees d) Animals
47. Which one is matching sanctuary
a) Kaziranga – musk deer b) Gir -Lion
c) N.E. Himalayan region – Samber d) Sunderban – rhino
48. Which group of vertebrates comprises the highest number of endangered species
a) Birds b) Mammals c) Fishes d) Reptiles
49. In our country the percentage of land under forest is about
a) 20% b) 19% c) 25% d) 30%
50. Fluoride pollution mainly affects
a) Kidney b) Brain c) Heart d) Teeth
51. The word Environment is derived from
a) Greek b) French c) Spanish d) English

- 52 Primary consumer
a) Herbivores b) Carnivores c) Macro consumer d) Omnivores
- 53 Abiotic component includes
a) Soil b) Temperature c) Water d) All the three
- 54 Atmosphere consists of 79% Nitrogen and 21% oxygen by
a) Volume b) Weight c) Density d) All the three
- 55 Which of the following is a biotic component of an eco system
a) Fungi b) Solar light c) Temperature d) Humidity
- 56 Which pyramid is always upright
a) Energy b) Biomass c) Number d) Food chain
- 57 Which of the following conceptual sphere of environment is having the least storage capacity for matter.
a) Atmosphere b) Lithosphere c) Hydrosphere d) Biosphere
- 58 The objective of environment education is
a) Raise consciousness about environmental conditions
b) To teach environmentally appropriate behavior
c) Create an environmental ethic.
d) All of the above
- 59 The sequence of eating and being eaten in an ecosystem is called
a) Food chain b) Carbon cycle
c) Hydrological cycle d) Anthroposystem
- 60 The largest reservoir of Nitrogen in our planet is
a) Ocean b) Atmosphere c) Biosphere d) Hydrosphere
- 61 In an ecosystem the flow of energy is
a) Bidirectional b) Cyclic c) Unidirectional d) Multidirectional
- 62 Estuary means
a) Rich in nutrients
b) End point of the river
c) Meeting place of river and sea
d) Treatment of water
- 63 What percentage of its geographical area of a country should be under forest cover?
a) 23% b) 43% c) 13% d) 33%
- 64 Mineral resources are
a) Renewable b) Available in plenty c) Non-renewable d) Equally distributed
- 65 Which of the following is considered as an alternate fuel
a) CNG b) Kerosene c) Coal d) Petrol
- 66 Chernobyl nuclear disaster occurred in the year
a) 1984 b) 1972 c) 1986 d) 1987
- 67 Electromagnetic radiation can cause
a) Plague b) Malaria c) Cancer d) Fever

- 68 OTEC is an energy technology that convert
 a) Energy in large tides ocean to generate electricity
 b) Energy in ocean wave to generate electricity
 c) Energy in ocean due to thermal gradient to generate electricity
 d) Energy in the fast moving ocean current to generate
- 69 Direct conversion of solar energy is attained by
 a) Solar photo voltaic system
 b) Solar hybrid system
 c) Solar thermal system
 d) Solar air heater
- 70 Hydrogen energy can be trapped through
 a) Heat pump
 b) Fuel cell
 c) Photovoltaic cell
 d) Cell
- 71 Effect of modern agriculture on soil is due to
 a) Erosion
 b) Acidification
 c) Salinization
 d) All
- 72 Bio – remediation means removal of containments from
 a) Soil
 b) Wastewater
 c) Ground water
 d) All
- 73 Population explosion will cause
 a) Bio diversity
 b) Stress in ecosystem
 c) Unemployment
 d) None of these
- 74 Smog is combination of
 a) Smoke and Snow
 b) Snow and Fog
 c) Smoke and Fog
 d) All the above
- 75 Physical pollution of water is due to
 a) pH
 b) Chloride
 c) Turbidity
 d) All of these
- 76 Renewable energy is
 a) Primary source
 b) Secondary source
 c) Tertiary source
 d) none of the above
- 77 Generation of wind energy is mainly based on
 a) Storm
 b) Pressure
 c) Velocity of wind
 d) Direction of wind
- 78 Which of the following is not a green house gas?
 a) Ozone
 b) Water vapour
 c) CO₂
 d) Sulpher dioxide
- 79 World ozone day is being celebrated on every year
 a) Sept 5th
 b) Sept 16th
 c) Oct 15th
 d) June 5th
- 80 First International earth summit was held at
 a) Germany
 b) Stock Holm
 c) Riode Janeiro
 d) England
- 81 Cholera and typhoid are caused by
 a) Worms
 b) Virus
 c) Bacteria
 d) Fungus

- 82 Hepatitis is caused by
a) Protozoa b) Virus c) Fungus d) Bacteria
- 83 The required iron content in drinking water as specified by BIS is
a) 300 mg/l b) 30 mg/l c) 3 mg/l d) 0.3 mg/l
- 84 The pH value of the acid rain water is less than
a) 3.2 b) 5.7 c) 7.0 d) 8.7
- 85 Reduction of brightness of famous Taj Mahal is due to
a) Global warming b) Air pollution c) Ozone depletion d) Afforestation
- 86 Ozone layer thickness is measured in
a) PPM b) PPB c) Decibels d) Dobson unit
- 87 Ferons are
a) HFC b) CFC c) NFC d) Hydrocarbon
- 88 Bhopal gas tragedy caused due to leakage of
a) Methyl iso cyanate (MIC)
b) Sulphur dioxide
c) Mustard gas
d) Chlorine gas
- 89 ICDS is a welfare scheme for
a) Public b) Women c) Men d) Children
- 90 Study of trends in human population growth and prediction of future growth is called
a) Demography b) Biography c) Kalography d) Psychology
- 91 The leader of Chipko movement is
a) Sunderlal Bahuguna b) Medha Patkar c) Vandana Shiva d) Anna Hazare
- 92 The Tiger conservation project was started in
a) 1973 b) 1984 c) 1999 d) 2004
- 93 An international conference on environmental education was held in December 1982 at
a) Kyoto b) Vienna c) New Delhi d) Rome
- 94 ISO 14000 standards deals with
a) Population management
b) Risk management
c) Environmental management
d) None of the above
- 95 The First International Earth Summit was held at
a) Johannesburg b) Rio de Janeiro c) Kyoto d) Stock holm
- 96 Disease caused by eating fish inhabiting mercury contaminated water is
a) Bright's disease
b) Minimata disease
c) Hashimoto disease
d) Osteosclerosis

- 97 Nitrogen oxide and hydrocarbons released by automobiles interact to form
a) Carbon monoxide b) Ammonia c) PAN d) Aerosols
- 98 The world AIDS day is recalled on
a) 1st July b) 5th June c) 1st December d) 5th October
- 99 ENVIS Means
a) Environmental Information System
b) Environmental Interference service
c) Environmental Information Service
d) None of the above
- 100 Corona viruses is a
a) DNA viruses
b) RNA viruses
c) Chromosomal viruses
d) None of the above

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CBCS SCHEME

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18CV71

Seventh Semester B.E. Degree Examination, Feb./Mar. 2022 Quantity Surveying and Contracts Management

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 What is an estimate? Explain briefly purpose and different types of estimates (any three). (20 Marks)

OR

- 2 The details of two room building are shown in the Fig.Q2. Estimate quantities and cost of the following items of work:
- (i) Earthwork excavation foundation at Rs.380/m³
 - (ii) Bed concrete 1 : 4 : 8 for foundation at Rs.2600/m³
 - (iii) S.S.M. for foundation and basement at Rs.3600/m³

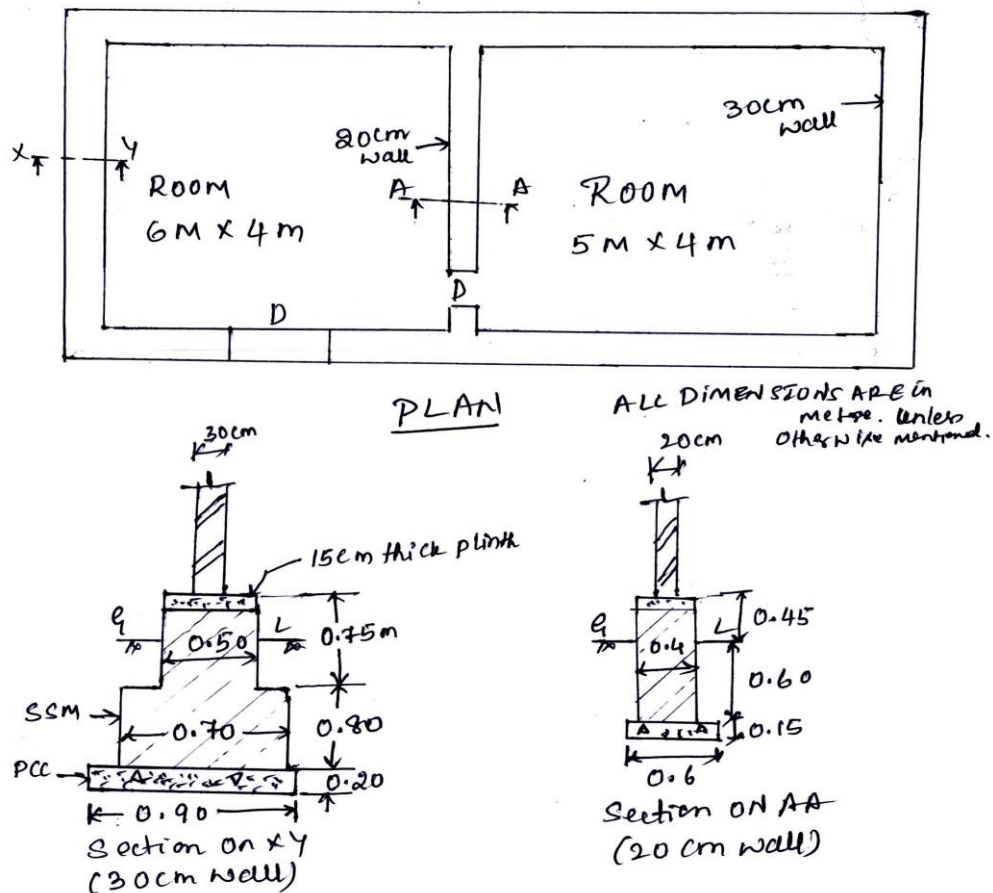


Fig.Q2

(20 Marks)

Module-2

- 3 The details of septic tank are shown in Fig.Q3. Estimate the quantities of following items and cost. (20 Marks)

- (i) Earthwork excavation at Rs. 400/m³
- (ii) PCC 1 : 3 : 6 for bed at Rs. 2500/m³
- (iii) BBM in CM 1 : 4 at Rs. 2200/m
- (iv) R.C.C. 1 : 2 : 4 roof slab cover at Rs. 3000/m³
- (v) 12mm cement plaster for sidewalls at Rs. 200/m²

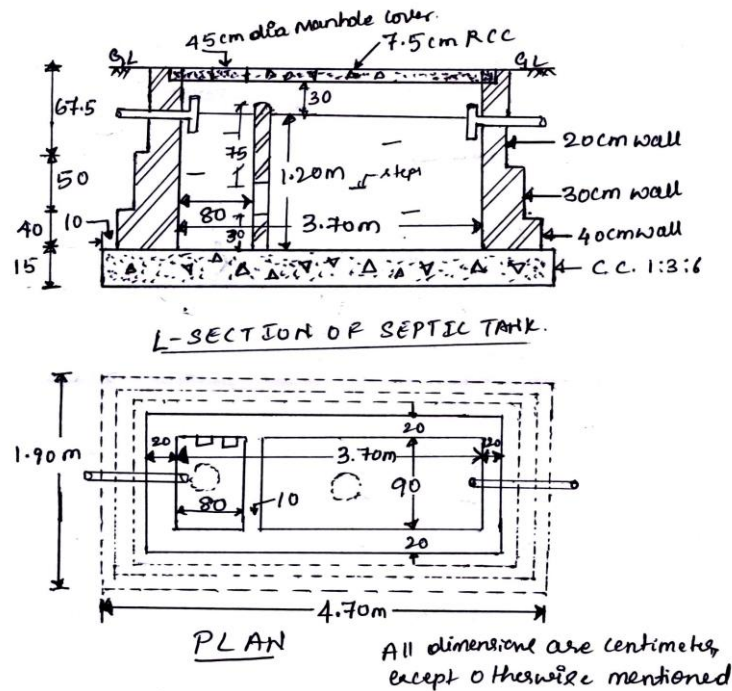


Fig.Q3

OR

- 4 Prepare a detailed estimate for earthwork for a portion of road from the following data:

Dist. in m	0	100	200	300	400	500	600	700	800	900	1000	1100	1200
RL of the ground	114.50	114.75	115.25	115.20	116.10	116.85	118.00	118.25	118.10	117.80	117.75	117.90	119.50
RL of formation	115												
Gradient	Upward gradient 1 in 200							Downward gradient 1 in 400					

Formation width of road is 10m. Side slope 2:1 in banking and 1½ : 1 in cutting. Calculate also the cost of this earthwork in banking and cutting; the rates are Rs. 275/m³ and Rs. 350/m³. Adopt Mid-Sectional area method. (20 Marks)

Module-3

- 5 Write detailed specification for following :
- (i) Earthwork excavation for foundation
 - (ii) Damp proof course 2.5cm (1") C.C. 1 : 1½ : 3
 - (iii) Burnt brick masonry for superstructure in CM 1:6
 - (iv) R.C.C. 1 : 2 : 4 for roof slab.

(20 Marks)

OR

- 6 Analyse rates from first principle for following :
(i) Cement concrete 1 : 5 : 10 in foundation.
(ii) Ist class brick work in super structure with CM 1:6
(iii) Coursed Rubbed stone masonry in CM 1:6 for super structure.
(iv) 12 mm thick internal plastering in CM 1:6 for brick walls. (20 Marks)

Module-4

- 7 What is tender? Explain the departmental procedure of tendering civil works. (20 Marks)

OR

- 8 What are the different types of contracts? Explain any four types of contracts. (20 Marks)

Module-5

- 9 Write a short notes on :
a. Mobilization and equipment advance
b. Secured advance
c. Liquidated damages and bonus
d. Dispute resolution mechanism
e. Performance security. (20 Marks)

OR

- 10 What is valuation? Explain briefly methods of valuation of buildings. (20 Marks)

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CBCS SCHEME

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18CV72

Seventh Semester B.E. Degree Examination, Feb./Mar. 2022 Design of RCC and Steel Structures

Time: 3 hrs.

Max. Marks:100

- Note:** 1. Answer any ONE full question each from Module-1 and Module-2.
2. Use of IS – 456, SP – 16, IS – 800, SP(6) and steel tables is permitted.
3. Missing data, if any, may be suitably assumed and same must be stated clearly.

Module - 1

- 1 Design slab type or slab beam type combined footing for two columns A and B spaced at 3.5m center to center. Cross section dimensions of column A is 400×400 mm and carries an axial load of 1050kN. Cross section dimensions of column B is 500×500 mm and carries an axial load of 1250kN. Safe bearing capacity of the foundation soil is 240 kN/m^2 . The width of the combined footing is restricted to 2.00m. Use M-25 grade concrete and Fe-415 grade steel. Draw neat sketch of reinforcement details. (50 Marks)

OR

- 2 Design A cantilever retaining wall to retain soil embankment for a height of 3.5m above the average ground level. The back fill is horizontal at the top. The unit weight of soil is 16 kN/m^3 and safe bearing capacity of the formation soil is 150 kN/m^2 . The angle of repose of the soil is 30° and the coefficient of friction between concrete surface and soil may be taken as 0.55. Use M – 20 grade concrete and Fe – 415 steel. Draw a neat sketch of the designed reinforcement details. (50 Marks)

Module – 2

- 3 Design a welded gantry girder to be used in an industrial building for carrying a manually operated over head crane for the data as listed below :
Crane capacity = 200kN
Self weight of crab consisting trolley, motor, hooks, etc. = 40kN
Self weight of crane girder excluding crab (trolley) = 200kN
Minimum hook approach = 1.20m
Wheel base of crab (trolley) = 3.50m
Centre to centre distance between gantry rails = 16m(span of crane girder)
Centre to centre distance between columns = 8.00 (span for gantry girder)
Self weight of rail section = 300N/m
Diameter of crane wheel = 150mm
The steel used is Fe – 410 grade. Draw a neat sketch of the designed details. (50 Marks)

4. Design a bolted steel Howe truss having an effective span of 12.00m. The geometry of the truss is as shown in Fig.Q4. The forces induced in the members due to dead load, live load and wind load is tabulated in Table.Q4. Determine the design forces in the members due to various combination of loads as per IS – 800 provisions and hence design principal rafter, principal tie and main sling member with all the necessary safety checks including the reversal of stresses. Also design support joint 'A' by considering the size of supporting reinforced cement concrete column as $300 \times 300\text{mm}$, the design bearing pressure on the concrete is limited to 2N/mm^2 and the design bond stress between anchor bolt and concrete is limited to 1.2N/mm^2 . Use M16 ordinary black bolts of grade 4.6 for designing member and connection with gusset plate and M25 bolt as anchor bolt at supports. List the design details.

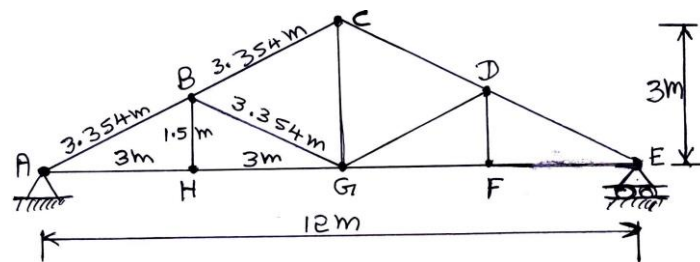


Fig.Q4

Members	Dead load kN	Live load kN	Wind load kN
Rafter AB, BC, CD and DE	-58.00	-52.52	+95.60
Tie member AH, HG, GF and FE	+52.00	+47.00	-76.00
Main sling BG, DG	+20.30	+18.40	-63.00

Table Q.4

Note :

- (-) indicates compressive force
- (+) indicates tensile force
- Net support reaction $\uparrow = 45\text{kN}$ (at 'A')
- Net up-lift support reaction $\downarrow = 55\text{kN}$ (At 'A').

(50 Marks)

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Seventh Semester B.E. Degree Examination, Feb./Mar. 2022

Air Pollution and Control

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Air Pollution. Explain primary and secondary Air pollution in detail. (10 Marks)
- b. With respect to Air pollution, explain Air borne contaminants. (04 Marks)
- c. Define the different sources of Air pollution and explain any 3 sources. (06 Marks)

OR

- 2 a. What are the effects of carbon dioxide on Plant, Animal and Human life? Explain. (10 Marks)
- b. Write a note on Photochemical Smog. (04 Marks)
- c. Define : i) Fog ii) Mist iii) Smoke iv) Soot v) Particulate vi) Droplet. (06 Marks)

Module-2

- 3 a. Explain in detail the temperature lapse rate with the neat sketch. (10 Marks)
- b. Explain in detail wind rose diagram and pollutant – wind correlation. (10 Marks)

OR

- 4 a. With the help of neat sketch, explain in detail the various types of Plume behaviour. (12 Marks)
- b. Explain the effects of Air pollutant on Meteorology. (08 Marks)

Module-3

- 5 a. What is meant by Air sampling? Explain Non – iso kinetic and Iso – kinetic sampling. (12 Marks)
- b. Write a note on Indoor Air Pollution. (08 Marks)

OR

- 6 a. Explain the Gaussian plume dispersion equation for the gaseous pollutants. (12 Marks)
- b. It is proposed to establish a new coal fired 1000MW power plant which works with 35% efficiency. If coal with 4% sulphur with a heat content of 14000 B_{tu/lb} emits SO₂ at the rate of 143000 lb/day from a stack with an effective height equal to 200m, estimate the ground level concentration of SO₂ at a downwind distance of 4km. The wind speed at 10m height is 3.5m/s and it is a cloudy summer day. Also estimate at what downwind distance maximum concentration of SO₂ can be observed. (08 Marks)

Module-4

- 7 a. With the help of neat sketch, explain the working principle of Electrostatic precipitator. (10 Marks)
- b. With the help of neat sketch, explain the working principle of wet scrubbers. (10 Marks)

OR

- 8 a. Explain the factors to be considered for the selection of Industrial plant location. (10 Marks)
b. with the help of neat sketch, explain the working principle of Fabric filters. (10 Marks)

Module-5

- 9 a. Define Noise pollution. Explain in detail the causes , effects and control of Noise pollution. (12 Marks)
b. Brief about Air pollution due to Automobiles. (08 Marks)

OR

- 10 a. Explain the Bhopal Gas Tragedy in detail. (10 Marks)
b. What is Green House effect? Explain briefly effect of Green house effect on Environment. (10 Marks)

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Seventh Semester B.E. Degree Examination, Feb./Mar. 2022

Pavement Materials and Construction

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Briefly explain the basic classification of aggregates. (05 Marks)
- b. Write a note on aggregate blending to meet the specified gradation. (05 Marks)
- c. Explain and indicate the suitability of following tests on the selection of proper aggregates for various pavement constructions with their permissible limits:
 - i) Aggregate impact test
 - ii) LOS Angles abrasion test. (10 Marks)

OR

- 2 a. Explain the manufacturing process of bitumen with a neat sketch. (10 Marks)
- b. Explain the following tests conducted on bituminous binders with neat sketches:
 - i) Penetration test
 - ii) Softening point test. (10 Marks)

Module-2

- 3 a. Explain briefly the preparation of emulsion and types of emulsions. (10 Marks)
- b. What are cutbacks? How are they prepared? Mention types of cutbacks with applications of each type. (10 Marks)

OR

- 4 a. What is stripping of bitumen? Briefly explain the mechanism of stripping and mention the methods of improving adhesion. (10 Marks)
- b. List out types of adhesion tests. Explain in brief about immersion trafficking test with neat diagram. (10 Marks)

Module-3

- 5 a. Explain the mechanical properties of Bituminous mix. (10 Marks)
- b. Explain briefly Marshall method of Bituminous mix design. (10 Marks)

OR

- 6 a. The specific gravities and weight proportions for aggregate and bitumen are as under for the preparation of Marshall mix design. The volume and weight of one Marshall mix design. The volume and weight of one Marshall specimen was found to be 475CC and 1100gm. Assuming absorption of bitumen in aggregate is zero, find V_v , V_b , VMA and VFB.

Item	Aggregate	Aggregate 2	Aggregate 3	Aggregate 4	Bitumen
Weight (gm)	825	1200	325	150	100
Special gravity	2.63	2.51	2.46	2.43	1.05

(12 Marks)

- b. Explain how the optimum bitumen content is determined. (08 Marks)

Module-4

- 7 a. With neat sketch, explain i) Dragline ii) Power shovel. (10 Marks)
 b. Discuss the suitability of different compacting equipments. Briefly explain working principle of sheeps foot rollers, with neat sketch. (10 Marks)

OR

- 8 a. Enumerate the steps involved in the formation of embankment. (10 Marks)
 b. Write short notes on:
 i) Desirable properties of subgrade soil
 ii) Quality control tests for subgrade construction. (10 Marks)

Module-5

- 9 a. Explain the material specification, construction procedure for water Bound Macadam roads. (10 Marks)
 b. Write a note on construction of Dense Bituminous Macadam pavement. (10 Marks)

OR

- 10 a. With the neat figures, explain the following types of joints in rigid pavements:
 i) Expansion joint
 ii) Contraction joint
 iii) Longitudinal joint. (12 Marks)
 b. Differentiate between flexible and rigid pavements. (08 Marks)

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Seventh Semester B.E. Degree Examination, Feb./Mar. 2022

Urban Transport Planning

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain system approach in Transportation Planning, with a neat flow diagram. (10 Marks)
- b. Explain the causes and effects of Urbanization. (10 Marks)

OR

- 2 a. Briefly explain the coordination required in Public transportation. (10 Marks)
- b. Write a note on : i) BRTS ii) Metro trains. (10 Marks)

Module-2

- 3 a. Define Zoning. Discuss the points to be kept in mind while doing Zoning. (10 Marks)
- b. What are the methods available for origin and destination study? Explain Home Interview method in detail. (10 Marks)

OR

- 4 a. Define Expansion factor. Explain briefly the accuracy checks necessary for the data collected by any survey. (10 Marks)
- b. List the different sampling techniques. Explain briefly. (10 Marks)

Module-3

- 5 a. Explain in detail the various factors governing trip generation. (10 Marks)
- b. What is Multiple Linear Regression Analysis? Mention the assumptions and limitations of Multiple Linear Regression Analysis. (10 Marks)

OR

- 6 a. Enlist the different methods of Trip distribution. Explain in detail Average Growth Factor method and Uniform Growth Factor method. (10 Marks)
- b. Estimate the future trip distribution as per Furness method [upto two iterations]. The predicted future trips are given in Table Q6(b) below.

O \ D	1	2	3	4	Predicted Future Trips
1	8	3	16	15	147
2	6	9	8	5	42
3	10	8	3	8	32
4	2	4	7	12	30
Predicted Future trips	39	24	68	120	

Table Q 6(b)

(10 Marks)

Module-4

- 7 a. Explain briefly types of Opportunity models. (10 Marks)
- b. The total number of trips produced in and attracted to three Zones A, B and C of a survey area in the design year are tabulated in Table Q7(b) below. It is known that the trips between two zones are inversely proportional to the second power of the travel time between zones which is uniformly 20 minutes. If the trip interchange between Zones B and C is known to be 600. Calculate the trip interchange between Zones A & B , A & C , B & A and C & B.

Zone	Trips produced	Trips attracted
A	2000	3000
B	3000	4000
C	4000	2000

Table Q7(b)

(10 Marks)

OR

- 8 a. Define Modal Split. Explain in brief the factors affecting modal split. (10 Marks)
- b. The number of trips produced in and attracted to the three Zones 1, 2 and 3 are tabulated in table Q8(b) (1) below. The order of closeness of Zones is given in table Q8(b)(2) and the Zonal 'L' factor is given in table Q8(b)(3). Distribute the trips between Zones.

Zone	1	2	3	Total
Trips produced	14	33	28	75
Trips attracted	33	28	14	75

Table Q8(b)(1)

O \ D	1	2	3
1	1	2	3
2	2	1	3
3	2	3	1

Table Q8(b)(2)

Zone	1	2	3
L factor	0.04	0.02	0.04

Table Q8(b)(3)

(10 Marks)

Module-5

- 9 a. List the various assignment techniques and explain any two methods in brief. (10 Marks)
- b. Write a flow chart of fundamental structure of Lowry model and explain the principal components of the model. (10 Marks)

OR

- 10 a. Explain Diversion Curves with limitations. (10 Marks)
- b. Define Trip assignment Explain the application, with a neat flow chart. (10 Marks)

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18CV752

Seventh Semester B.E. Degree Examination, Feb./Mar.2022

Numerical Methods and Applications

Time: 3 hrs.

Max. Marks: 100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Missing data may be suitably assumed.**

Module-1

- 1 a. Find the real root of the equation, $\cos x = 3x - 1$ correct to three decimal places using iteration method. (10 Marks)
- b. Solve the system of linear equations by using Gauss elimination method,
- $$4x_1 + 2x_2 + 3x_3 = 4$$
- $$2x_1 + 2x_2 + x_3 = 6$$
- $$x_1 - x_2 + x_3 = 0$$
- (10 Marks)

OR

- 2 a. Using Newton-Raphson method, find the real root of $x \log_{10} x = 1.2$ correct to five decimal places. (10 Marks)
- b. Solve the following set of linear equations by Gauss-Seidel method.
- $$10x_1 + x_2 + x_3 = 12$$
- $$2x_1 + 10x_2 + x_3 = 13$$
- $$2x_1 + 2x_2 + 10x_3 = 14$$
- Carryout five iterations. (10 Marks)

Module-2

- 3 a. Using Lagrange's interpolation formula, find a polynomial which passes through the points (0, -12), (1, 0), (3, 6), (4, 12). (05 Marks)
- b. Using Lagrange's interpolation formula, find the value of 'y' corresponding to $x = 10$ from the following table: (05 Marks)

x	5	6	9	11
y = f(x)	12	13	14	16

- c. The values of $\sin x$ are given below for different values of x . Find the value of $\sin 32^\circ$ using Newton's forward interpolation formula. (10 Marks)

x =	30°	35°	40°	45°	50°
y = sin x	0.5	0.5736	0.6428	0.7071	0.7660

OR

- 4 a. Use Newton's divided difference formula and evaluate $f(6)$ given,

x	5	7	11	13	21
f(x)	150	392	1452	2366	9702

(10 Marks)

- b. The following data gives the melting point of an alloy of lead and zinc, where t is the temperature in degrees C and P is the percentage of lead in the alloy. Find the melting point of the alloy containing 84% lead. Use Newton's backward difference formula.

P	40	50	60	70	80	90
t	184	204	226	250	276	304

(10 Marks)

Module-3

- 5 a. Find the value of $\int_0^1 \frac{dx}{1+x^2}$, taking 5 sub intervals by trapezoidal rule. Correct to five significant figures. Also compare it with its exact value. (10 Marks)
- b. Evaluate $\int_2^3 \frac{dx}{1+\sin x}$ by using Gauss quadrature three point formula. (10 Marks)

OR

- 6 a. The velocity of a train which starts from rest is given by the following table Table Q6 (a). Estimate approximately the total distance run in 20 minutes using Simpson's $\frac{1}{3}$ rule.

(10 Marks)

t(min)	2	4	6	8	10	12	14	16	18	20
v(km / hr)	16	28.8	40	46.4	51.2	32	17.6	8	3.2	0

Table Q6 (a)

- b. Evaluate the integral $I = \int_0^{0.2} \int_0^{0.2} e^y \sin x \, dx \, dy$ by ,

- (i) Trapezoidal rule with $h = k = 0.2$ and
- (ii) Simpson's $\frac{1}{3}$ rule with $h = k = 0.1$

(10 Marks)

Module-4

- 7 a. Find by Taylor's series method, the values of y at $x = 0.1$ and $x = 0.2$ to five places of decimals from $\frac{dy}{dx} = x^2y - 1$, $y(0) = 1$. (10 Marks)
- b. Given $\frac{dy}{dx} = \frac{1}{2}(1+x^2)y^2$ and $y(0) = 1$, $y(0.1) = 1.06$, $y(0.2) = 1.12$, $y(0.3) = 1.21$. Evaluate $y(0.4)$ by Milne's predictor corrector method. (10 Marks)

OR

- 8 a. Solve by Euler's method the following differential equation at $x = 0.1$, correct to four decimal places, with the initial condition $y(0) = 1$, $h = 0.02$, $\frac{dy}{dx} = \frac{y-x}{y+x}$. (10 Marks)
- b. Using Runge-Kutta method of order 4, find 'y' for $x = 0.1, 0.2$ given that $\frac{dy}{dx} = xy + y^2$, $y(0) = 1$. (10 Marks)

Module-5

- 9 The deflection of a beam is governed by the equation $\frac{d^4y}{dx^4} + 81y = \phi(x)$, where $\phi(x)$ is given by the table,

x	$\frac{1}{3}$	$\frac{2}{3}$	1
$\phi(x)$	81	162	243

and boundary condition $y(0) = y'(0) = y'(1) = y''(1) = 0$. Evaluate the deflection at the pivoted points of the beam using three sub-intervals. **(20 Marks)**

OR

- 10 a. Solve the equation $\frac{\partial u}{\partial y} = \frac{\partial^2 u}{\partial x^2}$ subject to the conditions $u(x,0) = \sin \pi x$, $0 \leq x \leq 1$; $u(0,t) = 0$, $u(1,t) = 0$, using Crank-Nicolson method. Carryout computations for two levels, taking $h = \frac{1}{3}$, $K = \frac{1}{36}$. **(10 Marks)**
- b. Find the solution of the initial boundary value problem, $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$, $0 \leq x \leq 1$; subject to the initial conditions $u(x,0) = \sin \pi x$, $0 \leq x \leq 1$, $\left(\frac{\partial u}{\partial x}\right)(x,0) = 0$, $0 \leq x \leq 1$ and the boundary conditions $u(0,t) = 0$, $u(1,t) = 0$, $t > 0$, by using in the explicit scheme. Take $h = K = 0.2$. **(10 Marks)**

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18CV753

Seventh Semester B.E. Degree Examination, Feb./Mar. 2022 Environmental Protection and Management

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Enumerate the various Environmental Management Principles and explain any two principles. (10 Marks)
- b. Enumerate the various drivers and barriers for sustainable development (any five in each). (10 Marks)

OR

- 2 a. Explain Abatement of Pollution. List out the major activities initiated under the various schemes on Pollution Abatement. (10 Marks)
- b. Enumerate the Business charter tools for sustainable production and consumption (any ten). (10 Marks)

Module-2

- 3 a. Enumerate the various Environmental Quality objectives. Briefly explain any two. (10 Marks)
- b. Distinguish between :
 - i) Pollution control and Pollution Prevention. (10 Marks)
 - ii) Concentration and Mass Standards. (10 Marks)

OR

- 4 a. Explain Environmental Performance Evaluation Process using PDCA Cycle. (10 Marks)
- b. Briefly explain :
 - i) Cleaner production and Clean technology. (10 Marks)
 - ii) Zero discharge technologies. (10 Marks)

Module-3

- 5 a. Enumerate the benefits and barriers in implementing ISO 14001 in an Organisation. (10 Marks)
- b. Explain EMS and EMAS. (10 Marks)

OR

- 6 a. Write Explanatory notes on :
 - i) Initial Environmental Review (IER) (10 Marks)
 - ii) Management Review (MR). (10 Marks)
- b. Write a note on Environmental aspect and Impacts within an Organisational setup. (10 Marks)

Module-4

- 7 a. Write a process flow diagram for the Management of an Audit Programme as per ISO - 19011. (12 Marks)
- b. Enumerate the contents of Environmental Statement (Form – V). (08 Marks)

OR

- 8** a. Write a note on Waste Minimisation Planning in an Industry. (10 Marks)
b. Explain briefly :
i) Non – conformity ii) Roles and Qualifications of Auditors. (10 Marks)

Module-5

- 9** a. Briefly discuss the applications of EMS and Waste Audit. (10 Marks)
b. Explain the concept of Transboundary of Pollutants. (10 Marks)

OR

- 10** a. Discuss Pollution Prevention Opportunities in Pulp Paper and Sugar mills. (10 Marks)
b. Write a note on Hazardous Waste classification and characteristics. (10 Marks)

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